

Appendix

Appendix A: Detailed results for MIP formulations with and without VIs

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Table A.1: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 0.6$, $RDD = 0.6$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	4	MIP-P	Optimal	0.09	282.78	0
		MIP-P+VI	Optimal	0.09	6.78	0
		MIP-T	Feasible	9.00	1800	100
		MIP-T+VI	Feasible	1.05	1800	100
1	3	MIP-P	Optimal	22.00	23.30	0
		MIP-P+VI	Optimal	22.00	1.68	0
		MIP-T	Optimal	22.00	264.39	0
		MIP-T+VI	Optimal	22.00	36.70	0
2	3	MIP-P	Optimal	43.14	14.53	0
		MIP-P+VI	Optimal	43.14	1.11	0
		MIP-T	Optimal	43.14	560.65	0
		MIP-T+VI	Optimal	43.14	74.08	0
3	4	MIP-P	Optimal	1.38	66.49	0
		MIP-P+VI	Optimal	1.38	6.64	0
		MIP-T	Optimal	1.38	1008.90	0
		MIP-T+VI	Optimal	1.38	742.99	0
4	4	MIP-P	Optimal	0.07	60.22	0
		MIP-P+VI	Optimal	0.07	9.26	0
		MIP-T	Feasible	0.18	1800	100
		MIP-T+VI	Feasible	0.07	1800	100
5	3	MIP-P	Optimal	1.41	38.23	0
		MIP-P+VI	Optimal	1.41	0.25	0
		MIP-T	Feasible	31.11	1800	100
		MIP-T+VI	Optimal	1.41	102.43	0
6	3	MIP-P	Optimal	2.27	27.11	0
		MIP-P+VI	Optimal	2.27	2.04	0
		MIP-T	Optimal	2.27	686.76	0
		MIP-T+VI	Optimal	2.27	119.99	0
7	3	MIP-P	Optimal	711.71	53.14	0
		MIP-P+VI	Optimal	711.71	0.24	0
		MIP-T	Optimal	711.71	787.53	0
		MIP-T+VI	Optimal	711.71	16.80	0
8	3	MIP-P	Optimal	49.82	39.66	0
		MIP-P+VI	Optimal	49.82	0.56	0
		MIP-T	Optimal	49.82	861.53	0
		MIP-T+VI	Optimal	49.82	26.74	0
9	4	MIP-P	Optimal	2.04	488.42	0
		MIP-P+VI	Optimal	2.04	3.35	0
		MIP-T	Feasible	2.04	1800	100
		MIP-T+VI	Optimal	2.04	1092.87	0
Avg	–	MIP-P	–	–	109.49	–
		MIP-P+VI	–	–	3.19	–
		MIP-T	–	–	1297.08	–
		MIP-T+VI	–	–	581.36	–

Note: A gap of 100% indicates that, at the time limit, the only available lower bound is the trivial nonnegativity lower bound of 0, rather than that no lower bound exists. The same convention applies to Tables A.2–A.9.

Table A.2: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 0.6$, $RDD = 0.8$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	4	MIP-P	Optimal	0.16	47.75	0
		MIP-P+VI	Optimal	0.16	6.38	0
		MIP-T	Feasible	2.46	1800	100
		MIP-T+VI	Optimal	0.16	1784.58	0
1	2	MIP-P	Optimal	58.98	1.66	0
		MIP-P+VI	Optimal	58.98	0.02	0
		MIP-T	Optimal	58.98	86.64	0
		MIP-T+VI	Optimal	58.98	1.81	0
2	3	MIP-P	Optimal	0.14	23.59	0
		MIP-P+VI	Optimal	0.14	9.16	0
		MIP-T	Feasible	0.14	1800	100
		MIP-T+VI	Optimal	0.14	220.24	0
3	3	MIP-P	Optimal	51.25	6.06	0
		MIP-P+VI	Optimal	51.25	0.06	0
		MIP-T	Optimal	51.25	77.94	0
		MIP-T+VI	Optimal	51.25	1.93	0
4	3	MIP-P	Optimal	0.25	48.67	0
		MIP-P+VI	Optimal	0.25	1.62	0
		MIP-T	Optimal	0.25	290.36	0
		MIP-T+VI	Optimal	0.25	153.77	0
5	4	MIP-P	Optimal	0.20	151.69	0
		MIP-P+VI	Optimal	0.20	5.18	0
		MIP-T	Feasible	0.41	1800	100
		MIP-T+VI	Feasible	0.20	1800	100
6	3	MIP-P	Optimal	13.96	10.25	0
		MIP-P+VI	Optimal	13.96	0.97	0
		MIP-T	Optimal	13.96	510.38	0
		MIP-T+VI	Optimal	13.96	71.55	0
7	4	MIP-P	Optimal	2.20	7.70	0
		MIP-P+VI	Optimal	2.20	0.37	0
		MIP-T	Feasible	2.20	1800	100
		MIP-T+VI	Optimal	2.20	157.00	0
8	4	MIP-P	Optimal	948.68	1.63	0
		MIP-P+VI	Optimal	948.68	0.12	0
		MIP-T	Optimal	948.68	274.14	0
		MIP-T+VI	Optimal	948.68	7.99	0
9	4	MIP-P	Optimal	0.07	124.79	0
		MIP-P+VI	Optimal	0.07	6.51	0
		MIP-T	Optimal	0.07	215.69	0
		MIP-T+VI	Optimal	0.07	155.04	0
Avg	–	MIP-P	–	–	42.38	–
		MIP-P+VI	–	–	3.04	–
		MIP-T	–	–	865.51	–
		MIP-T+VI	–	–	435.39	–

Table A.3: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 0.6$, $RDD = 1.0$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	3	MIP-P	Optimal	15.14	4.21	0
		MIP-P+VI	Optimal	15.14	1.31	0
		MIP-T	Optimal	15.14	1194.02	0
		MIP-T+VI	Optimal	15.14	166.46	0
1	1	MIP-P	Optimal	473.14	0.69	0
		MIP-P+VI	Optimal	473.14	0.01	0
		MIP-T	Optimal	473.14	187.25	0
		MIP-T+VI	Optimal	473.14	1.13	0
2	2	MIP-P	Optimal	1.75	2.88	0
		MIP-P+VI	Optimal	1.75	1.63	0
		MIP-T	Optimal	1.75	1174.82	0
		MIP-T+VI	Optimal	1.75	1506.14	0
3	3	MIP-P	Optimal	7.88	16.40	0
		MIP-P+VI	Optimal	7.88	0.20	0
		MIP-T	Optimal	7.88	85.89	0
		MIP-T+VI	Optimal	7.88	20.16	0
4	2	MIP-P	Optimal	169.20	18.05	0
		MIP-P+VI	Optimal	169.20	0.40	0
		MIP-T	Optimal	169.20	200.56	0
		MIP-T+VI	Optimal	169.20	3.95	0
5	2	MIP-P	Optimal	145.07	5.53	0
		MIP-P+VI	Optimal	145.07	0.26	0
		MIP-T	Optimal	145.07	471.85	0
		MIP-T+VI	Optimal	145.07	33.47	0
6	2	MIP-P	Optimal	450.89	6.13	0
		MIP-P+VI	Optimal	450.89	1.12	0
		MIP-T	Optimal	450.89	643.77	0
		MIP-T+VI	Optimal	450.89	93.39	0
7	3	MIP-P	Optimal	2.95	2.81	0
		MIP-P+VI	Optimal	2.95	1.42	0
		MIP-T	Optimal	2.95	192.53	0
		MIP-T+VI	Optimal	2.95	47.54	0
8	3	MIP-P	Optimal	177.52	2.45	0
		MIP-P+VI	Optimal	177.52	0.53	0
		MIP-T	Optimal	177.52	267.15	0
		MIP-T+VI	Optimal	177.52	16.45	0
9	4	MIP-P	Optimal	0.07	118.52	0
		MIP-P+VI	Optimal	0.07	6.53	0
		MIP-T	Optimal	0.07	214.15	0
		MIP-T+VI	Optimal	0.07	160.83	0
Avg	–	MIP-P	–	–	17.77	–
		MIP-P+VI	–	–	1.34	–
		MIP-T	–	–	463.20	–
		MIP-T+VI	–	–	204.95	–

Table A.4: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 0.8$, $RDD = 0.6$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	7	MIP-P	Feasible	0.02	1800	100
		MIP-P+VI	Optimal	0	83.95	0
		MIP-T	Feasible	0.04	1800	100
		MIP-T+VI	Optimal	0	1197.60	0
1	5	MIP-P	Optimal	3.27	88.65	0
		MIP-P+VI	Optimal	3.27	0.77	0
		MIP-T	Optimal	3.27	843.52	0
		MIP-T+VI	Optimal	3.27	111.43	0
2	6	MIP-P	Optimal	0	102.46	0
		MIP-P+VI	Optimal	0	8.58	0
		MIP-T	Feasible	0.12	1800	100
		MIP-T+VI	Optimal	0	85.83	0
3	7	MIP-P	Optimal	0	1226.00	0
		MIP-P+VI	Optimal	0	210.49	0
		MIP-T	Feasible	0.02	1800	100
		MIP-T+VI	Feasible	0.07	1800	100
4	6	MIP-P	Feasible	0.02	1800	100
		MIP-P+VI	Optimal	0.02	4.79	0
		MIP-T	Feasible	0.23	1800	100
		MIP-T+VI	Optimal	0.02	512.44	0
5	6	MIP-P	Optimal	17.68	562.92	0
		MIP-P+VI	Optimal	17.68	3.42	0
		MIP-T	Feasible	100.05	1800	100
		MIP-T+VI	Optimal	17.68	263.20	0
6	6	MIP-P	Feasible	0.05	1800	100
		MIP-P+VI	Optimal	0.05	35.09	0
		MIP-T	Feasible	0.39	1800	100
		MIP-T+VI	Feasible	0.34	1800	100
7	6	MIP-P	Feasible	0.20	1800	100
		MIP-P+VI	Optimal	0.20	6.57	0
		MIP-T	Feasible	1.20	1800	100
		MIP-T+VI	Optimal	0.20	1746.13	0
8	6	MIP-P	Optimal	0.04	26.99	0
		MIP-P+VI	Optimal	0.04	6.63	0
		MIP-T	Feasible	0.86	1800	100
		MIP-T+VI	Optimal	0.04	1282.29	0
9	6	MIP-P	Optimal	0.04	1023.23	0
		MIP-P+VI	Optimal	0.04	4.99	0
		MIP-T	Feasible	0.20	1800	100
		MIP-T+VI	Feasible	0.04	1800	100
Avg	–	MIP-P	–	–	1023.02	–
		MIP-P+VI	–	–	36.53	–
		MIP-T	–	–	1704.35	–
		MIP-T+VI	–	–	1059.89	–

Table A.5: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 0.8$, $RDD = 0.8$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	5	MIP-P	Optimal	1.29	26.69	0
		MIP-P+VI	Optimal	1.29	4.98	0
		MIP-T	Feasible	2.16	1800	100
		MIP-T+VI	Optimal	1.29	716.75	0
1	4	MIP-P	Optimal	3.82	4.03	0
		MIP-P+VI	Optimal	3.82	0.50	0
		MIP-T	Optimal	3.82	435.11	0
		MIP-T+VI	Optimal	3.82	14.36	0
2	5	MIP-P	Optimal	0	28.23	0
		MIP-P+VI	Optimal	0	2.90	0
		MIP-T	Optimal	0	280.09	0
		MIP-T+VI	Optimal	0	170.64	0
3	6	MIP-P	Optimal	0.02	664.80	0
		MIP-P+VI	Optimal	0.02	126.20	0
		MIP-T	Feasible	0.02	1800	100
		MIP-T+VI	Feasible	0.02	1800	100
4	5	MIP-P	Feasible	0.14	1800	100
		MIP-P+VI	Optimal	0.14	13.72	0
		MIP-T	Feasible	0.57	1800	100
		MIP-T+VI	Feasible	0.57	1800	100
5	5	MIP-P	Optimal	0.09	512.25	0
		MIP-P+VI	Optimal	0.09	21.85	0
		MIP-T	Feasible	0.23	1800	100
		MIP-T+VI	Feasible	0.23	1800	100
6	4	MIP-P	Optimal	26.98	17.20	0
		MIP-P+VI	Optimal	26.98	0.63	0
		MIP-T	Optimal	26.98	817.65	0
		MIP-T+VI	Optimal	26.98	81.28	0
7	5	MIP-P	Feasible	0.41	1800	100
		MIP-P+VI	Optimal	0.41	5.76	0
		MIP-T	Optimal	0.41	1791.42	0
		MIP-T+VI	Optimal	0.41	380.24	0
8	5	MIP-P	Optimal	0.04	136.66	0
		MIP-P+VI	Optimal	0.04	8.86	0
		MIP-T	Feasible	0.04	1800	100
		MIP-T+VI	Optimal	0.04	1316.77	0
9	5	MIP-P	Optimal	420.41	24.69	0
		MIP-P+VI	Optimal	420.41	0.25	0
		MIP-T	Optimal	420.41	243.50	0
		MIP-T+VI	Optimal	420.41	8.34	0
Avg	–	MIP-P	–	–	501.46	–
		MIP-P+VI	–	–	18.57	–
		MIP-T	–	–	1256.78	–
		MIP-T+VI	–	–	808.84	–

Table A.6: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 0.8$, $RDD = 1.0$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	4	MIP-P	Optimal	0.16	51.44	0
		MIP-P+VI	Optimal	0.16	6.68	0
		MIP-T	Feasible	2.46	1800	100
		MIP-T+VI	Optimal	0.16	1668.75	0
1	4	MIP-P	Optimal	0.48	14.96	0
		MIP-P+VI	Optimal	0.48	4.36	0
		MIP-T	Optimal	0.48	800.69	0
		MIP-T+VI	Optimal	0.48	322.67	0
2	4	MIP-P	Optimal	0.07	32.10	0
		MIP-P+VI	Optimal	0.07	4.87	0
		MIP-T	Optimal	0.07	1704.55	0
		MIP-T+VI	Feasible	0.07	1800	100
3	4	MIP-P	Optimal	5.38	20.24	0
		MIP-P+VI	Optimal	5.38	2.42	0
		MIP-T	Optimal	5.38	561.62	0
		MIP-T+VI	Optimal	5.38	94.42	0
4	4	MIP-P	Optimal	1.82	141.55	0
		MIP-P+VI	Optimal	1.82	3.13	0
		MIP-T	Optimal	1.82	840.91	0
		MIP-T+VI	Optimal	1.82	121.17	0
5	4	MIP-P	Optimal	0.20	94.96	0
		MIP-P+VI	Optimal	0.20	2.21	0
		MIP-T	Feasible	5.46	1800	100
		MIP-T+VI	Optimal	0.20	800.62	0
6	4	MIP-P	Optimal	0.88	65.79	0
		MIP-P+VI	Optimal	0.88	6.63	0
		MIP-T	Feasible	3.64	1800	100
		MIP-T+VI	Feasible	0.88	1800	100
7	5	MIP-P	Optimal	0.30	237.66	0
		MIP-P+VI	Optimal	0.30	11.72	0
		MIP-T	Optimal	0.30	591.30	0
		MIP-T+VI	Optimal	0.30	1143.26	0
8	4	MIP-P	Optimal	16.89	4.36	0
		MIP-P+VI	Optimal	16.89	0.21	0
		MIP-T	Optimal	16.89	570.71	0
		MIP-T+VI	Optimal	16.89	70.63	0
9	4	MIP-P	Optimal	559.45	98.00	0
		MIP-P+VI	Optimal	559.45	0.15	0
		MIP-T	Optimal	559.45	647.99	0
		MIP-T+VI	Optimal	559.45	23.73	0
Avg	–	MIP-P	–	–	76.11	–
		MIP-P+VI	–	–	4.24	–
		MIP-T	–	–	1111.78	–
		MIP-T+VI	–	–	784.52	–

Table A.7: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 1.0$, $RDD = 0.6$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	9	MIP-P	Optimal	0	102.22	0
		MIP-P+VI	Optimal	0	9.11	0
		MIP-T	Feasible	0.02	1800	100
		MIP-T+VI	Feasible	0.04	1800	100
1	8	MIP-P	Optimal	0	0.59	0
		MIP-P+VI	Optimal	0	13.01	0
		MIP-T	Feasible	0.73	1800	100
		MIP-T+VI	Feasible	0.07	1800	100
2	10	MIP-P	Optimal	0	46.00	0
		MIP-P+VI	Optimal	0	4.18	0
		MIP-T	Optimal	0	928.25	0
		MIP-T+VI	Feasible	0.02	1800	100
3	10	MIP-P	Optimal	0	39.80	0
		MIP-P+VI	Optimal	0	11.40	0
		MIP-T	Optimal	0	596.44	0
		MIP-T+VI	Feasible	0.02	1800	100
4	8	MIP-P	Optimal	0	1620.94	0
		MIP-P+VI	Optimal	0	2.53	0
		MIP-T	Optimal	0	365.07	0
		MIP-T+VI	Optimal	0	219.41	0
5	8	MIP-P	NoSolve	-	1800	0
		MIP-P+VI	Optimal	0	7.15	0
		MIP-T	Feasible	3.68	1800	100
		MIP-T+VI	Feasible	0.02	1800	100
6	9	MIP-P	Optimal	0	98.25	0
		MIP-P+VI	Optimal	0	45.45	0
		MIP-T	Feasible	0.04	1800	100
		MIP-T+VI	Feasible	0.02	1800	100
7	8	MIP-P	Optimal	0	221.44	0
		MIP-P+VI	Optimal	0	68.45	0
		MIP-T	Feasible	1.18	1800	100
		MIP-T+VI	Feasible	0.07	1800	100
8	9	MIP-P	Optimal	0	256.30	0
		MIP-P+VI	Optimal	0	25.34	0
		MIP-T	Optimal	0	543.69	0
		MIP-T+VI	Feasible	0.04	1800	100
9	8	MIP-P	Feasible	0.02	1800	100
		MIP-P+VI	Optimal	0.02	56.34	0
		MIP-T	Feasible	0.09	1800	100
		MIP-T+VI	Feasible	0.09	1800	100
Avg	-	MIP-P	-	-	598.55	-
		MIP-P+VI	-	-	24.30	-
		MIP-T	-	-	1323.35	-
		MIP-T+VI	-	-	1641.94	-

Table A.8: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 1.0$, $RDD = 0.8$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	8	MIP-P	Optimal	0	12.11	0
		MIP-P+VI	Optimal	0	6.04	0
		MIP-T	Feasible	0.04	1800	100
		MIP-T+VI	Optimal	0	500.99	0
1	7	MIP-P	Optimal	0	546.92	0
		MIP-P+VI	Optimal	0	23.61	0
		MIP-T	Feasible	0.41	1800	100
		MIP-T+VI	Optimal	0	1231.47	0
2	6	MIP-P	Optimal	0.05	180.21	0
		MIP-P+VI	Optimal	0.05	7.31	0
		MIP-T	Feasible	1.68	1800	100
		MIP-T+VI	Optimal	0.05	712.93	0
3	7	MIP-P	Optimal	0	125.77	0
		MIP-P+VI	Optimal	0	105.47	0
		MIP-T	Optimal	0	1703.77	0
		MIP-T+VI	Feasible	0.02	1800	100
4	7	MIP-P	Feasible	0.18	1800	100
		MIP-P+VI	Optimal	0.18	5.32	0
		MIP-T	Feasible	0.18	1800	100
		MIP-T+VI	Optimal	0.18	266.59	0
5	7	MIP-P	Optimal	147.07	1153.78	0
		MIP-P+VI	Optimal	147.07	2.09	0
		MIP-T	Feasible	258.70	1800	100
		MIP-T+VI	Optimal	147.07	88.17	0
6	7	MIP-P	Optimal	0	214.63	0
		MIP-P+VI	Optimal	0	29.96	0
		MIP-T	Feasible	5.41	1800	100
		MIP-T+VI	Feasible	0.52	1800	100
7	7	MIP-P	Feasible	0.02	1800	100
		MIP-P+VI	Optimal	0.02	3.17	0
		MIP-T	Optimal	0.02	730.02	0
		MIP-T+VI	Optimal	0.02	911.42	0
8	7	MIP-P	Feasible	0.04	1800	100
		MIP-P+VI	Optimal	0.02	36.90	0
		MIP-T	Feasible	0.02	1800	100
		MIP-T+VI	Feasible	0.04	1800	100
9	7	MIP-P	Feasible	0.04	1800	100
		MIP-P+VI	Optimal	0	7.13	0
		MIP-T	Feasible	0.05	1800	100
		MIP-T+VI	Optimal	0	775.59	0
Avg	–	MIP-P	–	–	943.34	–
		MIP-P+VI	–	–	22.70	–
		MIP-T	–	–	1683.38	–
		MIP-T+VI	–	–	988.72	–

Table A.9: Performance of MIP formulations with and without VIs ($n = 15$, $TF = 1.0$, $RDD = 1.0$, time limit = 1800 s)

Inst.#	$U_{\min}$	Method	Status	η	Time (s)	Gap (%)
0	7	MIP-P	Feasible	0.02	1800	100
		MIP-P+VI	Optimal	0	82.45	0
		MIP-T	Feasible	0.04	1800	100
		MIP-T+VI	Optimal	0	1151.53	0
1	5	MIP-P	Optimal	3.27	89.59	0
		MIP-P+VI	Optimal	3.27	0.78	0
		MIP-T	Optimal	3.27	801.17	0
		MIP-T+VI	Optimal	3.27	108.78	0
2	6	MIP-P	Optimal	0	102.72	0
		MIP-P+VI	Optimal	0	8.65	0
		MIP-T	Feasible	0.12	1800	100
		MIP-T+VI	Optimal	0	82.56	0
3	7	MIP-P	Optimal	0	1186.47	0
		MIP-P+VI	Optimal	0	202.79	0
		MIP-T	Feasible	0.02	1800	100
		MIP-T+VI	Feasible	0.07	1800	100
4	6	MIP-P	Feasible	0.02	1800	100
		MIP-P+VI	Optimal	0.02	4.86	0
		MIP-T	Feasible	0.23	1800	100
		MIP-T+VI	Optimal	0.02	510.54	0
5	6	MIP-P	Optimal	17.68	567.30	0
		MIP-P+VI	Optimal	17.68	3.46	0
		MIP-T	Feasible	100.05	1800	100
		MIP-T+VI	Optimal	17.68	264.11	0
6	6	MIP-P	Feasible	0.05	1800	100
		MIP-P+VI	Optimal	0.05	35.33	0
		MIP-T	Feasible	0.39	1800	100
		MIP-T+VI	Feasible	0.34	1800	100
7	6	MIP-P	Feasible	0.20	1800	100
		MIP-P+VI	Optimal	0.20	6.67	0
		MIP-T	Feasible	1.20	1800	100
		MIP-T+VI	Optimal	0.20	1751.25	0
8	6	MIP-P	Optimal	0.04	27.24	0
		MIP-P+VI	Optimal	0.04	6.68	0
		MIP-T	Feasible	0.86	1800	100
		MIP-T+VI	Optimal	0.04	1277.51	0
9	6	MIP-P	Optimal	0.04	1022.04	0
		MIP-P+VI	Optimal	0.04	4.99	0
		MIP-T	Feasible	0.20	1800	100
		MIP-T+VI	Feasible	0.04	1800	100
Avg	–	MIP-P	–	–	1019.54	–
		MIP-P+VI	–	–	35.67	–
		MIP-T	–	–	1700.12	–
		MIP-T+VI	–	–	1054.63	–

Table B.1: MIP-P+VI results for $(TF, RDD) = (0.6, 0.6)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	135.30	3.32	0.30	23.71	0	57.49	0	86.12	0	80.21	0	12.72
	1	329.30	0.59	0	3289.98	0	36.45	0	17.52	0	10.84	0	15.13
	2	1442.70	11.52	357.20	183.85	0	68.54	0	450.19	0	23.94	0	13.13
	3	0	14.03	0	1420.92	0	123.33	0	23.99	0	2.85	0	10.49
	4	0.10	28.50	0	149.92	0	5.48	0	18.62	0	0.98	0	1.10
	5	434.80	3.44	0	66.81	0	2.88	0	47.83	0	16.56	0	1.54
	6	0.50	2.30	0	18.38	0	198.50	0	75.23	0	13.96	0	15.65
	7	1.00	10.42	0	68.54	0	29.65	0	11.82	0	0.20	0	0.15
	8	0	58.71	0	17.61	0	6.18	0	61.58	0	12.56	0	39.56
	9	119.60	2.38	0	45.20	0	9.75	0	10.44	0	1.57	0	9.81
	Avg.	246.33	13.52	35.75	528.49	0	53.83	0	80.33	0	16.37	0	11.93
30	0	0	76.15	–	TL	0	2047.73	0	27.49	0	298.66	0	42.81
	1	0.07	51.34	0	85.08	0	22.70	0	58.26	–	TL	0	93.21
	2	0	97.88	0	79.89	0	1471.12	0	255.11	0	10.27	0	30.23
	3	0	83.34	–	TL	–	TL	–	TL	0	276.85	0	41.20
	4	0.40	27.57	0	352.89	0	3.46	–	TL	0	27.93	0	22.85
	5	–	TL	–	TL	–	TL	0	100.35	0	34.88	0	45.28
	6	0.07	67.77	0	1798.93	0	178.45	0	141.00	0	44.94	0	52.86
	7	0.80	19.30	–	TL	–	TL	0	313.63	0	1562.35	0	24.99
	8	0	103.68	0	138.82	0	582.52	0	173.50	0	39.28	0	75.04
	9	0	23.78	–	TL	0	630.90	–	TL	0	204.99	0	2.14
	Avg.	0.15	415.08	0	2045.56	0	1573.69	0	1186.93	0	610.02	0	43.06

Notes: “TL” indicates that the 3600-second time limit was reached. An underlined η indicates that MIP-P+VI returned a feasible incumbent at the time limit without proving optimality. “–” indicates that no feasible solution was obtained within the time limit. In each average row, $\bar{\eta}$ is computed over the instances with an available feasible solution, while $\bar{t}$ is computed over all 10 instances, with “TL” counted as 3600 s. The same convention applies to Tables B.2–B.9.

Table B.2: MIP-P+VI results for $(TF, RDD) = (0.6, 0.8)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	925.89	0	0.19	0	5.18	0	96.21	0	4.91	0	4.03
	1	8.90	2.16	0	8.85	0	6.16	0	19.68	0	3.01	0	10.89
	2	207.90	3.64	0	5.22	0	5.20	0	13.73	0	14.08	0	2.90
	3	0	11.08	0	68.38	0	2276.75	0	14.75	0	1.31	0	5.74
	4	0.10	46.10	0	12.14	0	1.19	0	8.43	0	6.58	0	1.60
	5	1.30	0.43	0	19.43	0	30.71	0	130.46	0	10.26	0	1.55
	6	0.50	15.80	0	5.38	0	9.92	0	0.90	0	9.07	0	9.99
	7	0.10	99.38	0	5.16	0	8.58	0	1.29	0	5.34	0	0.70
	8	0	33.44	0	3.37	0	11.63	0	43.17	0	19.67	0	0.36
	9	1.90	6.87	0	12.48	0	6.05	0	3.15	0	10.82	0	7.94
	Avg.	22.07	114.48	0	14.06	0	236.14	0	33.18	0	8.50	0	4.57
30	0	0.60	47.98	–	TL	0	219.24	0	76.11	0	0.44	0	9.07
	1	0.87	35.01	0	3147.42	0	75.33	0	73.12	0	44.36	0	40.64
	2	44.80	15.54	–	TL	0	55.63	0	40.88	0	65.84	0	61.91
	3	<u>0.07</u>	TL	–	TL	0	911.24	0	537.18	0	36.21	0	18.52
	4	<u>0.07</u>	TL	–	TL	0	1491.73	0	42.29	0	1.97	0	30.18
	5	–	TL	0	75.05	0	95.81	0	352.09	0	39.63	–	TL
	6	0	371.14	0	68.55	0	3130.27	0	14.07	0	25.10	0	1.49
	7	1.33	12.79	0	169.36	0	76.55	–	TL	0	60.11	0	40.77
	8	0.07	76.61	–	TL	0	219.53	–	TL	0	582.27	0	2.26
	9	–	TL	0	1133.73	0	1598.48	0	65.98	0	257.07	0	29.57
	Avg.	5.98	1495.91	0	2259.41	0	787.38	0	840.17	0	111.30	0	383.44

Table B.3: MIP-P+VI results for $(TF, RDD) = (0.6, 1.0)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	1.60	13.45	0	13.49	0	26.86	0	3.75	0	9.71	0	11.22
	1	73.20	0.61	0	5.06	0	242.05	0	26.39	0	17.03	0	48.88
	2	0	384.02	0	13.69	0	70.91	0	331.56	0	7.81	0	0.45
	3	46.90	0.89	0	23.01	0	9.12	0	0.67	0	27.24	0	1.29
	4	0.10	22.91	0	7.66	0	27.71	0	1.21	0	18.12	0	17.93
	5	235.90	0.18	0	202.95	0	53.23	0	12.41	0	9.49	0	10.45
	6	0	0.61	0	6.32	0	0.74	0	10.97	0	0.02	0	1.26
	7	0.30	18.25	0	56.25	0	17.10	0	8.92	0	7.64	0	1.81
	8	0	13.02	0	8.77	0	1.11	0	3.86	0	14.63	0	10.31
	9	6.60	0.13	0.10	2346.19	0	44.09	0	36.93	0	9.90	0	2.91
	Avg.	36.46	45.41	0.01	268.34	0	49.29	0	43.67	0	12.16	0	10.65
30	0	0.20	21.98	–	TL	0	650.56	0	1638.33	0	27.55	0	36.22
	1	243.07	50.94	0	69.80	0	761.75	0	15.40	0	46.89	0	28.35
	2	0.07	956.39	0	48.73	0	32.57	0	44.13	0	21.63	0	1.57
	3	3.20	30.05	0	140.28	0	24.15	0	2.97	0	27.32	0	20.43
	4	0	416.58	0	91.40	0	27.68	0	119.64	0	27.42	0	22.77
	5	0.07	TL	0	30.13	0	36.75	0	21.28	0	3.64	0	27.40
	6	578.73	1971.49	0	237.44	0	12.78	0	38.79	0	25.43	0	30.59
	7	0.67	0.74	0	14.48	0	21.44	0	28.68	0	19.39	0	14.82
	8	3.20	11.42	–	TL	0	27.73	0	2.72	0	12.27	0	24.90
	9	–	TL	0	58.74	0	39.37	0	28.37	0	108.42	0	1.50
	Avg.	92.13	1065.96	0	789.10	0	163.48	0	194.03	0	32.00	0	20.85

Table B.4: MIP-P+VI results for $(TF, RDD) = (0.8, 0.6)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	7.30	0	11.64	0	25.72	0	6.79	0	12.06	0	8.42
	1	0	25.86	0	31.87	0	13.24	0	28.90	0	16.73	0	8.07
	2	0	19.15	0	41.42	0	15.55	0	9.29	0	11.81	0	23.70
	3	0	65.55	0	9.97	0	13.13	0	29.02	0	20.33	0	13.54
	4	0	31.37	0	5.52	0	11.05	0	6.30	0	9.35	0	11.26
	5	0	6.03	0	22.80	0	1.05	0	15.06	0	0.73	0	8.58
	6	0	36.15	0	41.57	0	12.07	0	9.11	0	20.24	0	0.13
	7	0	10.30	0	11.62	0	3.69	0	16.39	0	16.44	0	17.41
	8	0	1.35	0	9.98	0	20.03	0	8.68	0	0.72	0	3.94
	9	0	9.93	0	21.42	0	16.81	0	23.13	0	14.70	0	9.37
	Avg.	0	21.30	0	20.78	0	13.23	0	15.27	0	12.31	0	10.44
30	0	0	25.42	0	166.46	0	192.41	0	511.29	0	24.37	0	44.63
	1	0	25.19	0	97.18	0	53.00	0	399.32	0	78.74	0	99.57
	2	0	35.42	0	80.94	0	1132.48	0	8.96	0	3023.25	0	36.30
	3	0	246.77	0	123.82	0	28.79	0	48.19	0	31.64	0	33.74
	4	0	73.04	0	102.55	0	1646.52	0	87.90	0	89.45	0	52.01
	5	0	65.30	0	195.92	0	49.75	0	152.79	0	47.59	0	35.19
	6	0	11.19	0	1333.34	0	42.45	0	75.49	0	934.78	0	121.74
	7	0	62.54	0	272.56	0	457.59	0	64.32	0	442.21	0	37.74
	8	0	21.25	0	86.26	0	2253.35	0	22.49	0	20.70	0	28.20
	9	0	151.11	–	TL	0	219.03	0	70.31	0	34.36	0	26.39
	Avg.	0	71.72	0	605.90	0	607.54	0	144.11	0	472.71	0	51.55

Table B.5: MIP-P+VI results for $(TF, RDD) = (0.8, 0.8)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	15.77	0	13.05	0	14.43	0	1.41	0	11.53	0	0.96
	1	0	22.26	0	60.75	0	23.26	0	25.48	0	4.57	0	2.97
	2	0.20	10.89	0	27.35	0	11.10	0	5.80	0	17.83	0	11.15
	3	0	10.23	0	20.04	0	12.15	0	13.77	0	26.80	0	7.72
	4	0	10.87	0	2.07	0	45.60	0	12.11	0	33.23	0	11.31
	5	367.50	11.56	0	542.94	0	16.26	0	13.40	0	34.03	0	9.59
	6	31.40	22.37	0	10.23	0	1.74	0	1.60	0	11.96	0	10.81
	7	0.50	9.52	0	12.51	0	18.26	0	14.98	0	5.18	0	33.13
	8	0.10	2.70	0	34.31	0	8.51	0	1.58	0	14.38	0	0.75
	9	0	18.14	0	1.00	0	8.39	0	1.03	0	34.36	0	0.56
	Avg.	39.97	13.43	0	72.42	0	15.97	0	9.11	0	19.39	0	8.90
30	0	0	274.90	0	46.62	0	446.81	0	49.06	0	236.55	0	19.22
	1	0	47.05	–	TL	0	25.75	0	833.98	0	68.61	0	82.97
	2	0	40.84	0	233.15	0	45.44	0	39.73	0	96.16	0	27.09
	3	476.13	47.21	0	623.22	–	TL	0	111.36	0	59.88	0	20.53
	4	0	115.86	0	261.65	–	TL	0	123.36	–	TL	0	54.57
	5	0	86.35	0	727.76	0	31.28	0	82.02	0	333.75	0	25.73
	6	0	27.77	–	TL	0	278.36	0	47.99	0	42.42	0	7.87
	7	0	80.46	0	49.50	0	601.27	0	26.30	0	67.66	0	102.74
	8	0	60.08	0	21.49	–	TL	0	241.95	0	41.42	0	112.38
	9	0	1095.94	0	42.20	0	190.69	0	105.78	0	12.01	0	22.91
	Avg.	47.61	187.65	0	920.56	0	1241.96	0	166.15	0	455.85	0	47.60

Table B.6: MIP-P+VI results for $(TF, RDD) = (0.8, 1.0)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	12.58	0	836.47	0	100.50	0	6.69	0	0.86	0	4.21
	1	0	7.29	0	35.29	0	11.76	0	12.25	0	13.46	0	13.56
	2	0	17.21	0	12.62	0	3.26	0	0.91	0	0.72	0	9.99
	3	0.60	13.18	0	622.90	0	16.97	0	14.54	0	13.52	0	19.08
	4	0.10	37.46	0	40.26	0	30.93	0	11.06	0	7.82	0	11.08
	5	885.00	0.24	0	104.67	0	6.72	0	7.87	0	8.15	0	2.69
	6	0.20	6.99	0	1.07	0	20.38	0	11.50	0	16.84	0	0.52
	7	0.20	9.75	0	6.75	0	42.68	0	25.07	0	8.55	0	8.02
	8	0.10	19.13	0	84.88	0	229.27	0	2.44	0	3.00	0	9.50
	9	241.20	0.48	0	25.47	0	20.78	0	9.83	0	13.31	0	8.77
	Avg.	112.74	12.43	0	177.04	0	48.33	0	10.22	0	8.62	0	8.74
30	0	1899.60	1.93	333.13	1329.06	0	1347.14	0	2639.44	0	704.94	0	309.16
	1	0.40	15.28	–	TL	0	299.70	0	56.12	0	81.70	0	30.39
	2	0	224.38	0	487.72	–	TL	–	TL	0	26.97	0	75.80
	3	670.47	88.28	–	TL	–	TL	–	TL	0	119.11	0	142.06
	4	0	28.47	–	TL	–	TL	0	29.14	0	6.40	0	21.87
	5	0	32.59	–	TL	0	893.20	0	106.35	0	29.57	0	47.43
	6	0.33	4.18	–	TL	0	397.08	0	136.84	0	38.92	0	2319.41
	7	0.07	11.97	0	178.55	0	1203.29	0	139.73	–	TL	0	433.82
	8	0	69.44	–	TL	–	TL	0	125.07	0	38.72	0	981.00
	9	0	29.36	–	TL	–	TL	0	31.70	0	31.85	0	103.47
	Avg.	257.09	50.59	111.04	2719.53	0	2214.04	0	1046.44	0	467.82	0	446.44

Table B.7: MIP-P+VI results for $(TF, RDD) = (1.0, 0.6)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	25.51	0	8.42	0	16.31	0	3.42	0	30.03	0	10.04
	1	0	11.71	0	14.82	0	17.92	0	11.17	0	31.39	0	5.65
	2	470.50	4.18	0	19.79	0	8.35	0	20.99	0	25.31	0	18.44
	3	0	4.38	0	3.23	0	14.79	0	22.35	0	17.70	0	17.97
	4	0	7.94	0	7.09	0	8.45	0	7.40	0	14.89	0	26.65
	5	0	1.98	0	15.27	0	30.25	0	16.94	0	14.70	0	8.00
	6	0	33.62	0	13.94	0	18.44	0	10.03	0	3.37	0	14.53
	7	0	21.35	0	12.71	0	25.87	0	15.63	0	11.63	0	1.20
	8	0	18.69	0	89.05	0	26.08	0	28.67	0	12.88	0	23.70
	9	0	19.54	0	48.36	0	11.74	0	12.21	0	20.87	0	19.72
	Avg.	47.05	14.89	0	23.27	0	17.82	0	14.88	0	18.28	0	14.59
30	0	0	123.57	0	79.09	0	48.23	0	204.67	0	64.90	0	31.90
	1	0	500.77	0	35.75	0	35.77	0	97.53	0	281.56	0	95.32
	2	0	57.68	0	71.00	0	166.99	0	27.73	0	190.78	0	32.90
	3	0	52.34	0	75.81	0	62.84	0	269.69	0	30.03	0	141.37
	4	0	130.04	0	256.54	0	104.45	0	451.31	0	105.28	0	558.20
	5	0	36.95	0	191.53	0	172.88	0	898.68	0	77.99	0	98.22
	6	0	55.68	0	52.49	0	42.02	0	112.69	0	66.73	0	125.20
	7	–	TL	–	TL	0	171.17	0	254.38	0	760.72	0	331.66
	8	0	71.61	0	115.92	0	40.87	0	34.37	0	10.53	0	32.67
	9	0	62.41	0	155.43	0	33.16	0	98.70	0	154.14	0	52.38
	Avg.	0	469.11	0	463.36	0	87.84	0	244.97	0	174.27	0	149.98

Table B.8: MIP-P+VI results for $(TF, RDD) = (1.0, 0.8)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	7.64	0	67.65	0	12.35	0	13.90	0	13.29	0	3.21
	1	0	5.43	0	9.37	0	0.63	0	14.88	0	6.52	0	13.17
	2	0	14.97	0	14.81	0	12.95	0	0.68	0	45.37	0	9.18
	3	0	10.37	0	10.56	0	3.78	0	17.86	0	4.66	0	9.27
	4	0	16.61	0	22.76	0	11.59	0	12.27	0	14.49	0	5.72
	5	0	6.89	0	10.94	0	7.32	0	6.10	0	8.07	0	0.22
	6	0	8.96	0	11.73	0	9.61	0	10.32	0	13.06	0	7.21
	7	0	6.14	0	11.16	0	53.10	0	3.36	0	14.25	0	23.18
	8	0	14.65	0	8.86	0	6.76	0	19.75	0	21.13	0	9.78
	9	0	9.46	0	31.51	0	7.72	0	22.98	0	0.59	0	14.30
	Avg.	0	10.11	0	19.94	0	12.58	0	12.21	0	14.14	0	9.52
30	0	0	20.57	0	38.34	0	76.81	0	46.75	0	94.62	0	51.04
	1	0	247.00	–	TL	0	86.88	0	200.86	0	37.71	0	45.95
	2	0	89.22	–	TL	0	613.12	0	32.04	0	75.27	0	49.98
	3	505.33	216.15	0	138.71	–	TL	0	148.80	0	26.99	0	125.98
	4	690.87	131.51	0	67.03	0	51.15	0	313.14	0	57.79	0	88.75
	5	0	73.85	0	77.05	0	132.72	0	131.07	0	53.92	0	54.19
	6	0	261.38	0	94.42	–	TL	0	51.06	0	41.71	0	44.47
	7	559.07	25.02	–	TL	0	67.13	0	52.98	–	TL	0	66.25
	8	0	59.01	0	166.49	0	157.97	0	43.05	0	140.28	0	61.65
	9	0	60.55	0	55.23	0	28.93	0	76.25	0	3.65	0	40.54
	Avg.	175.53	118.43	0	1143.73	0	841.47	0	109.60	0	413.19	0	62.88

Table B.9: MIP-P+VI results for $(TF, RDD) = (1.0, 1.0)$: objective value η and computation time under different δ values

n	Inst.	$\delta = 0$		$\delta = 1$		$\delta = 2$		$\delta = 3$		$\delta = 4$		$\delta = 5$	
		η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)	η	t (s)
20	0	0	0.90	0	16.00	0	12.68	0	4.38	0	24.13	0	4.18
	1	0	26.71	0	9.59	0	19.54	0	1.98	0	0.82	0	4.50
	2	546.20	4.23	0	439.21	0	52.63	0	39.22	0	10.88	0	11.55
	3	0	33.65	0	1.12	0	4.83	0	1.93	0	10.70	0	1.18
	4	0	28.18	0	94.36	0	7.65	0	1.53	0	0.96	0	8.48
	5	0	15.52	0	14.07	0	26.74	0	35.31	0	12.55	0	17.91
	6	166.20	4.73	0	11.54	0	13.02	0	11.95	0	21.04	0	13.02
	7	0	11.70	0	270.78	0	33.09	0	13.89	0	9.75	0	4.27
	8	0	34.61	0	18.83	0	22.25	0	23.39	0	11.41	0	18.97
	9	0	5.39	0	16.33	0	0.45	0	0.70	0	8.86	0	10.71
	Avg.	71.24	16.56	0	89.18	0	19.29	0	13.43	0	11.11	0	9.48
30	0	0	2.41	—	TL	0	45.49	0	143.40	0	82.17	0	181.98
	1	0	44.81	—	TL	0	103.65	0	241.64	0	2.79	0	19.74
	2	0	68.11	0	413.36	0	344.41	0	428.45	0	39.83	0	18.09
	3	0	25.63	0	551.18	—	TL	0	55.36	—	TL	0	110.94
	4	0	46.61	0	221.88	0	27.14	0	24.36	0	107.54	0	35.90
	5	0	27.88	0	1470.30	0	292.71	0	57.51	0	52.11	0	26.19
	6	0	53.61	—	TL	0	97.33	0	70.27	0	89.30	0	6.57
	7	0	114.32	—	TL	0	69.07	0	224.62	0	70.15	0	25.56
	8	0	27.51	0	6.02	0	129.29	0	138.27	0	17.96	0	2.37
	9	0	70.62	0	248.12	0	1801.27	0	263.91	0	47.58	0	726.88
	Avg.	0	48.15	0	1731.09	0	651.04	0	164.78	0	410.94	0	115.42

Table C.1: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 0.6$, $RDD = 0.6$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	4	181.30	0.50	11.88	0	0.50	5.46	0.50	7.01
	1	4	975.70	7.30	5.42	0	10.00	3.65	7.30	5.98
	2	4	767.80	372.10	2.24	0	372.10	0.38	372.10	1.31
	3	5	1020.50	0.10	20.19	0	0.10	12.74	0.10	19.75
	4	5	1186.70	0	16.67	0	0	1.73	0	1.73
	5	5	374.30	0.30	11.35	0	0.30	11.29	0.30	24.80
	6	5	339.40	0.20	21.11	0	0.20	13.65	0.20	33.72
	7	4	195.20	0.30	5.12	0	0.30	2.39	0.30	6.68
	8	5	623.70	1.10	10.99	0	1.10	0.39	1.10	1.40
	9	5	512.60	0.30	16.83	0	0.30	3.83	0.30	22.34
30	0	6	74.40	0	88.51	0	0.07	3.07	0	285.25
	1	6	48.40	0	21.25	0	0	0.28	0	0.28
	2	5	1113.60	557.80	9.29	0	557.80	2.94	557.80	8.34
	3	5	431.00	0.40	TL	100	0.07	4.32	0.07	58.23
	4	6	460.13	72.00	6.73	0	72.00	0.07	72.00	3.44
	5	5	115.53	0.27	7.69	0	3.20	0.06	0.27	4.15
	6	5	201.13	0.13	58.48	0	0.20	4.32	0.13	76.44
	7	7	472.87	0	90.21	0	0	0.30	0	0.30
	8	6	623.00	0.27	140.43	0	0.27	0.71	0.27	60.20
	9	6	301.40	0	225.15	0	0	7.15	0	7.15
50	0	10	1050.16	0	3290.27	0	0	0.72	0	0.72
	1	9	850.44	0	285.20	0	0	0.27	0	0.27
	2	9	823.36	0.04	TL	100	0	0.47	0	0.47
	3	10	1268.40	0.08	TL	100	0	0.62	0	0.62
	4	10	466.24	0.04	TL	100	0	0.45	0	0.45
	5	9	1052.36	0	2171.54	0	0	0.23	0	0.23
	6	10	942.96	0.16	TL	100	0	0.72	0	0.72
	7	10	686.52	0	1409.54	0	0	1.04	0	1.04
	8	8	103.68	0.08	TL	100	0.04	1.13	0	1921.02
	9	9	757.92	0	905.57	0	0	0.85	0	0.85
80	0	13	199.70	–	–	–	0	0.56	0	0.56
	1	15	715.50	–	–	–	0	3.28	0	3.28
	2	13	469.63	–	–	–	0	0.73	0	0.73
	3	13	2328.38	–	–	–	0	1.66	0	1.66
	4	15	623.25	–	–	–	0	3.64	0	3.64
	5	16	585.50	–	–	–	0	0.78	0	0.78
	6	15	1115.42	–	–	–	0	1.58	0	1.58
	7	13	554.95	–	–	–	0	0.45	0	0.45
	8	14	466.80	–	–	–	0	1.46	0	1.46
	9	16	2653.60	–	–	–	0	0.86	0	0.86
100	0	17	621.68	–	–	–	0	0.35	0	0.35
	1	16	873.26	–	–	–	0	0.50	0	0.50
	2	20	3300.06	–	–	–	0	0.54	0	0.54
	3	17	2278.20	–	–	–	<u>4.80</u>	0.40	<u>4.80</u>	TL
	4	19	1388.00	–	–	–	0	0.93	0	0.93
	5	18	998.80	–	–	–	0	0.61	0	0.61
	6	18	2246.74	–	–	–	<u>4.80</u>	0.28	<u>4.80</u>	TL
	7	18	1900.78	–	–	–	0	0.78	0	0.78
	8	19	285.88	–	–	–	0	0.81	0	0.81
	9	17	490.44	–	–	–	<u>4.80</u>	0.11	<u>4.80</u>	TL

Note: “TL” indicates that the 3600-second time limit was reached. “NoFeasSol” indicates that no feasible solution was found within the time limit. A dash indicates that the corresponding method was not tested. For MIP-P+VI and CP, TL cases are counted as 3600 s when computing average time. Underlined CP objective values indicate heuristic incumbents returned at the time limit without an optimality certificate. The same convention applies to Tables C.2–C.9.

Table C.2: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 0.6$, $RDD = 0.8$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	3	683.20	423.10	3.96	0	423.10	0.70	423.10	0.77
	1	4	2086.40	2086.40	0.54	0	2086.40	0.02	2086.40	0.25
	2	3	289.90	0.10	12.97	0	115.00	3.44	0.10	4.77
	3	4	1460.50	406.70	9.90	0	406.70	4.62	406.70	5.30
	4	4	83.70	0	8.10	0	38.90	10.68	0	12.73
	5	4	755.70	3.20	19.53	0	4.70	2.25	3.20	4.44
	6	3	633.70	3.20	5.41	0	3.20	11.76	3.20	12.30
	7	3	78.20	1.90	18.47	0	1.90	37.03	1.90	38.41
	8	5	929.00	0.20	50.42	0	16.90	0.63	0.20	18.17
	9	5	490.10	0.10	403.45	0	66.70	0.74	0.10	88.76
30	0	4	268.40	62.87	8.49	0	62.87	3.41	62.87	5.14
	1	5	523.73	0	20.82	0	0	2.35	0	2.35
	2	4	748.27	360.53	30.19	0	360.53	0.86	360.53	3.63
	3	5	349.67	0.27	242.41	0	0.27	3.46	0.27	50.98
	4	5	532.53	0	91.70	0	0	2.28	0	2.28
	5	4	159.33	0.47	54.05	0	0.47	5.52	0.47	37.77
	6	4	210.00	0.07	1273.01	0	0.20	22.42	0.07	106.51
	7	5	1414.87	0	66.80	0	0	1.69	0	1.69
	8	7	746.27	0	27.91	0	0	0.75	0	0.75
	9	7	1229.73	0	9.82	0	5.33	0.05	0	11.11
50	0	8	1273.32	0.16	TL	100	0	2.10	0	2.10
	1	7	435.28	0.20	TL	100	0	0.48	0	0.48
	2	9	351.16	0.28	TL	100	0	0.16	0	0.16
	3	7	1036.80	355.96	71.46	0	420.08	0.17	355.96	186.12
	4	10	1156.48	0.08	TL	100	0	0.64	0	0.64
	5	9	661.12	0.16	TL	100	0	0.34	0	0.34
	6	9	654.08	NoFeasSol	TL	-	0	0.23	0	0.23
	7	9	656.32	268.36	TL	100	0	0.24	0	0.24
	8	7	208.16	0	2843.90	0	0.48	0.39	<u>0.04</u>	TL
	9	7	1575.20	0.04	TL	100	0	0.56	0	0.56
80	0	10	635.60	-	-	-	<u>344.85</u>	0.09	<u>344.85</u>	TL
	1	11	466.40	-	-	-	0	0.61	0	0.61
	2	10	256.15	-	-	-	0	0.90	0	0.90
	3	13	2982.62	-	-	-	0	0.73	0	0.73
	4	13	667.25	-	-	-	0	0.82	0	0.82
	5	13	968.63	-	-	-	0	0.88	0	0.88
	6	10	1485.37	-	-	-	<u>900.95</u>	0.11	<u>900.95</u>	TL
	7	9	585.20	-	-	-	0	0.94	0	0.94
	8	9	639.63	-	-	-	0	0.56	0	0.56
	9	15	3762.60	-	-	-	0	0.67	0	0.67
100	0	13	100.90	-	-	-	0	1.33	0	1.33
	1	11	892.58	-	-	-	<u>0.02</u>	0.51	<u>0.02</u>	TL
	2	13	502.02	-	-	-	0	0.71	0	0.71
	3	13	879.64	-	-	-	0	2.37	0	2.37
	4	13	509.78	-	-	-	0	0.43	0	0.43
	5	13	324.78	-	-	-	0	0.95	0	0.95
	6	12	1780.12	-	-	-	0	1.58	0	1.58
	7	17	845.82	-	-	-	0	0.82	0	0.82
	8	14	2855.42	-	-	-	0	1.85	0	1.85
	9	13	2140.70	-	-	-	0	0.79	0	0.79

Table C.3: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 0.6$, $RDD = 1.0$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	4	248.10	0	47.24	0	0.20	13.32	0	175.24
	1	3	925.00	310.40	4.15	0	841.60	0.12	310.40	0.25
	2	3	289.90	0.10	12.95	0	115.00	3.42	0.10	4.75
	3	3	564.00	275.10	8.50	0	383.10	0.59	275.10	2.03
	4	3	311.20	159.50	6.21	0	159.50	13.78	159.50	14.26
	5	2	966.80	966.80	0.85	0	966.80	0.03	966.80	0.05
	6	2	303.50	0.10	13.93	0	0.10	38.16	0.10	38.57
	7	2	23.00	0.90	4.36	0	0.90	13.83	0.90	13.94
	8	4	416.20	91.00	0.80	0	109.00	0.07	91.00	0.72
	9	4	2239.40	788.30	4.27	0	1205.30	0.46	788.30	0.82
30	0	2	67.27	5.13	12.68	0	5.13	0.30	5.13	0.38
	1	4	20.73	0.07	296.20	0	0.07	30.42	0.07	177.21
	2	2	1336.00	1336.00	7.47	0	1336.00	0.12	1336.00	1.07
	3	3	471.80	0.13	28.62	0	198.87	0.18	0.13	10.78
	4	4	277.60	0	239.68	0	0	2.72	0	2.72
	5	3	331.13	0.07	1219.47	0	2.07	24.23	0.07	71.95
	6	2	435.00	23.80	9.29	0	23.80	0.69	23.80	1.29
	7	3	335.27	0.07	TL	100	24.20	4.41	0.07	14.63
	8	6	455.20	76.40	70.26	0	165.40	0.05	76.40	66.38
	9	6	282.13	0	38.98	0	0.13	4.67	<u>0</u>	TL
50	0	6	862.44	0.24	TL	100	0	4.84	0	4.84
	1	5	660.08	NoFeasSol	TL	-	256.28	0.22	248.92	322.65
	2	6	539.80	NoFeasSol	TL	-	0	1.61	0	1.61
	3	5	574.08	0.08	TL	100	2.32	0.29	0.08	107.60
	4	3	1620.84	1620.80	6.67	0	1620.80	0.05	1620.80	7.89
	5	4	974.56	126.16	55.78	0	126.16	0.79	126.16	360.82
	6	5	14.24	0.04	TL	100	<u>0.08</u>	2.40	<u>0.08</u>	TL
	7	5	51.64	NoFeasSol	TL	-	0.12	4.79	<u>0.04</u>	TL
	8	5	142.56	0.44	TL	100	0.20	0.65	<u>0.04</u>	TL
	9	4	673.76	0.08	TL	100	<u>0.52</u>	10.32	<u>0.52</u>	TL
80	0	5	853.90	-	-	-	<u>609.60</u>	0.29	<u>609.60</u>	TL
	1	8	150.03	-	-	-	<u>1.33</u>	0.06	<u>1.33</u>	TL
	2	8	642.28	-	-	-	0	0.64	0	0.64
	3	9	1636.88	-	-	-	<u>0.30</u>	0.21	<u>0.30</u>	TL
	4	5	640.18	-	-	-	0.10	1.71	<u>0.05</u>	TL
	5	7	1338.17	-	-	-	0	0.98	0	0.98
	6	6	2252.68	-	-	-	<u>0.05</u>	5.92	<u>0.05</u>	TL
	7	5	625.08	-	-	-	<u>0.65</u>	0.40	<u>0.65</u>	TL
	8	6	316.23	-	-	-	0	1.17	0	1.17
	9	11	1058.07	-	-	-	<u>655.95</u>	0.09	<u>655.95</u>	TL
100	0	7	2186.16	-	-	-	<u>0.32</u>	0.33	<u>0.32</u>	TL
	1	6	136.64	-	-	-	0	3.55	0	3.55
	2	9	1930.78	-	-	-	0	1.03	0	1.03
	3	9	449.82	-	-	-	0	2.02	0	2.02
	4	7	211.66	-	-	-	0	4.13	0	4.13
	5	7	356.26	-	-	-	0	2.68	0	2.68
	6	8	742.32	-	-	-	0	1.55	0	1.55
	7	12	806.66	-	-	-	<u>0.02</u>	0.60	<u>0.02</u>	TL
	8	7	2353.96	-	-	-	0	3.64	0	3.64
	9	8	2025.54	-	-	-	0	2.41	0	2.41

Table C.4: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 0.8$, $RDD = 0.6$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	8	675.50	0	17.32	0	0	0.72	0	0.72
	1	8	55.10	0	7.45	0	0	0.33	0	0.33
	2	7	680.60	0.10	37.90	0	0.10	0.71	0.10	13.77
	3	7	1177.80	618.90	5.72	0	618.90	0.80	618.90	2.25
	4	8	556.80	0	12.77	0	0	0.66	0	0.66
	5	7	996.50	681.50	0.90	0	681.50	0.02	681.50	0.40
	6	8	1537.70	133.10	9.30	0	133.10	0.42	133.10	6.85
	7	7	80.00	0	19.01	0	0	0.38	0	0.38
	8	8	1069.50	0.80	2.44	0	122.10	0.02	0.80	1.26
	9	8	1681.10	674.00	6.13	0	674.00	1.08	674.00	8.69
30	0	9	667.40	0	8.31	0	0	0.01	0	0.01
	1	11	616.20	0	67.50	0	0	0.26	0	0.26
	2	10	2227.00	0	237.94	0	0	3.05	0	3.05
	3	10	350.07	0	21.02	0	0	0.46	0	0.46
	4	10	951.80	0	137.32	0	0	0.13	0	0.13
	5	11	535.20	0	63.65	0	0	0.25	0	0.25
	6	9	569.87	0	17.37	0	0	0.28	0	0.28
	7	11	84.40	0	80.69	0	0	0.10	0	0.10
	8	12	75.27	0	12.96	0	0	0.34	0	0.34
	9	12	85.33	0	231.34	0	0	0.38	0	0.38
50	0	17	764.44	0	3250.07	0	0	0.25	0	0.25
	1	16	291.16	0	94.82	0	0	0.10	0	0.10
	2	16	970.40	NoFeasSol	TL	-	0	0.15	0	0.15
	3	19	1461.32	NoFeasSol	TL	-	0	0.31	0	0.31
	4	17	1114.56	0.04	TL	100	0	0.54	0	0.54
	5	16	37.28	NoFeasSol	TL	-	0	0.32	0	0.32
	6	17	111.88	0	221.06	0	0	0.01	0	0.01
	7	16	477.40	0	3312.18	0	0	0.50	0	0.50
	8	15	445.92	0	3293.76	0	0	0.13	0	0.13
	9	17	834.56	0.04	TL	100	0	0.41	0	0.41
80	0	24	618.85	-	-	-	0	0.18	0	0.18
	1	26	894.53	-	-	-	0	0.36	0	0.36
	2	23	633.88	-	-	-	0	0.14	0	0.14
	3	24	509.95	-	-	-	0	0.33	0	0.33
	4	26	670.40	-	-	-	0	0.48	0	0.48
	5	26	2411.48	-	-	-	0	0.77	0	0.77
	6	27	2148.10	-	-	-	0	0.70	0	0.70
	7	25	201.58	-	-	-	0	0.01	0	0.01
	8	25	2731.07	-	-	-	0	2.27	0	2.27
	9	26	1935.57	-	-	-	0	0.21	0	0.21
100	0	31	274.52	-	-	-	0	0.87	0	0.87
	1	29	13.56	-	-	-	0	0.01	0	0.01
	2	31	1190.14	-	-	-	0	0.22	0	0.22
	3	32	2263.20	-	-	-	0	0.01	0	0.01
	4	37	585.62	-	-	-	0	0.35	0	0.35
	5	33	1704.50	-	-	-	0	0.51	0	0.51
	6	32	3667.08	-	-	-	<u>595.90</u>	0.11	<u>595.90</u>	TL
	7	32	554.86	-	-	-	0	0.21	0	0.21
	8	34	1173.60	-	-	-	0	0.22	0	0.22
	9	33	2019.08	-	-	-	<u>275.78</u>	0.08	<u>275.78</u>	TL

Table C.5: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 0.8$, $RDD = 0.8$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	6	22.30	0	21.70	0	0	0.48	0	0.48
	1	7	1039.90	0	117.78	0	0	0.62	0	0.62
	2	6	181.00	0	21.13	0	0.10	0.74	0	17.48
	3	7	1901.50	543.60	20.67	0	543.60	5.89	543.60	22.27
	4	7	556.50	0	12.93	0	0.10	0.25	0	15.68
	5	6	736.60	332.80	1.88	0	332.80	0.18	332.80	0.87
	6	6	702.00	0.40	8.35	0	0.40	0.41	0.40	2.27
	7	6	257.50	0	10.45	0	0.20	0.50	0	4.83
	8	7	61.20	0.10	224.62	0	0.10	1.06	0.10	30.29
	9	7	511.20	195.90	23.65	0	411.40	2.41	195.90	14.61
30	0	8	736.27	0	227.49	0	0	0.42	0	0.42
	1	9	1342.80	0	141.45	0	0	0.56	0	0.56
	2	7	1127.13	524.67	9.77	0	524.67	0.46	524.67	9.98
	3	7	201.33	0	16.18	0	0	0.59	0	0.59
	4	8	165.53	0	36.08	0	0.67	0.06	0	0.95
	5	8	324.20	0	395.48	0	0	0.53	0	0.53
	6	7	197.80	0	66.34	0	0	0.55	0	0.55
	7	10	257.07	0	366.40	0	0	0.68	0	0.68
	8	9	622.27	0	34.38	0	0	0.82	0	0.82
	9	9	1030.47	0	116.86	0	0	0.51	0	0.51
50	0	13	1247.36	734.64	2798.86	0	734.64	0.08	734.64	148.97
	1	13	41.52	0	2018.97	0	0	0.48	0	0.48
	2	13	764.80	0	3047.48	0	0	0.36	0	0.36
	3	14	1267.96	0.12	TL	100	0	0.58	0	0.58
	4	14	616.84	0	822.67	0	0	1.45	0	1.45
	5	13	1201.80	NoFeasSol	TL	-	0	0.28	0	0.28
	6	14	749.32	0.04	TL	100	0	0.33	0	0.33
	7	14	202.52	0	3260.03	0	0	0.20	0	0.20
	8	12	424.20	0	935.22	0	0	0.45	0	0.45
	9	13	131.44	0.08	TL	100	0	0.31	0	0.31
80	0	18	13.53	-	-	-	0	0.01	0	0.01
	1	21	989.75	-	-	-	0	0.91	0	0.91
	2	18	837.68	-	-	-	0	0.30	0	0.30
	3	18	2566.25	-	-	-	<u>640.13</u>	0.26	<u>640.13</u>	TL
	4	22	109.38	-	-	-	0	0.35	0	0.35
	5	22	1471.95	-	-	-	0	0.87	0	0.87
	6	21	3097.38	-	-	-	0	0.53	0	0.53
	7	18	869.40	-	-	-	0	0.01	0	0.01
	8	19	1565.22	-	-	-	0	0.01	0	0.01
	9	21	3040.85	-	-	-	<u>0.03</u>	0.24	<u>0.03</u>	TL
100	0	24	684.30	-	-	-	0	0.01	0	0.01
	1	22	662.36	-	-	-	0	0.21	0	0.21
	2	27	3377.48	-	-	-	<u>0.02</u>	0.26	<u>0.02</u>	TL
	3	24	1141.92	-	-	-	0	0.26	0	0.26
	4	27	709.78	-	-	-	<u>0.02</u>	0.13	<u>0.02</u>	TL
	5	24	270.38	-	-	-	<u>0.06</u>	0.18	<u>0.06</u>	TL
	6	25	3451.60	-	-	-	<u>1336.14</u>	0.09	<u>1336.14</u>	TL
	7	25	250.18	-	-	-	0	0.16	0	0.16
	8	25	296.80	-	-	-	0	0.17	0	0.17
	9	24	1895.72	-	-	-	<u>0.06</u>	0.19	<u>0.06</u>	TL

Table C.6: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 0.8$, $RDD = 1.0$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	4	510.30	0.80	6.16	0	0.80	1.61	0.80	3.07
	1	5	263.20	25.20	5.96	0	25.20	0.66	25.20	1.76
	2	4	730.60	2.10	8.75	0	2.10	0.73	2.10	2.15
	3	5	909.10	159.60	9.33	0	200.20	2.28	159.60	4.70
	4	5	377.80	3.20	7.36	0	3.20	9.15	3.20	10.95
	5	5	354.90	0	8.33	0	0	0.72	0	0.72
	6	5	1337.50	0	138.59	0	0	3.04	0	3.04
	7	5	310.70	0	12.88	0	0	1.19	0	1.19
	8	7	445.10	0	20.61	0	0	1.47	0	1.47
	9	5	1616.40	1516.70	0.63	0	1516.70	0.02	1516.70	0.21
30	0	6	711.40	0	175.31	0	0	0.63	0	0.63
	1	7	660.40	0.07	TL	100	0	1.40	0	1.40
	2	6	1268.07	0	702.65	0	0	0.80	0	0.80
	3	5	174.80	0	20.81	0	14.80	0.10	0	4.38
	4	7	16.27	0	100.53	0	0	0.59	0	0.59
	5	5	710.53	467.73	4.78	0	467.73	0.04	467.73	1.29
	6	6	261.93	0	145.08	0	0	1.65	0	1.65
	7	7	552.87	0	728.39	0	0	1.07	0	1.07
	8	7	249.13	0	227.02	0	0	0.96	0	0.96
	9	8	1376.93	0	780.70	0	0	0.85	0	0.85
50	0	10	1232.44	0	2327.36	0	0	0.63	0	0.63
	1	10	683.28	0.04	TL	100	0	1.14	0	1.14
	2	11	817.24	NoFeasSol	TL	-	0	0.35	0	0.35
	3	10	1211.76	149.80	TL	19.3	149.80	0.17	149.80	173.33
	4	12	545.28	0	1307.67	0	0.08	0.05	0.00	829.14
	5	10	663.48	NoFeasSol	TL	-	0	0.32	0	0.32
	6	12	1378.44	NoFeasSol	TL	-	0	0.29	0	0.29
	7	12	325.52	0	839.43	0	0	0.30	0	0.30
	8	9	673.56	NoFeasSol	TL	-	0	0.50	0	0.50
	9	11	1914.20	401.96	TL	60.8	455.96	0.06	401.96	1712.37
80	0	13	565.70	—	—	—	<u>0.05</u>	0.09	<u>0.05</u>	TL
	1	15	1057.62	—	—	—	0	0.25	0	0.25
	2	13	1075.60	—	—	—	0	0.39	0	0.39
	3	14	3211.32	—	—	—	<u>1812.58</u>	0.49	<u>1812.58</u>	TL
	4	15	547.88	—	—	—	0	0.27	0	0.27
	5	17	658.75	—	—	—	0	1.75	0	1.75
	6	15	2258.15	—	—	—	<u>1465.75</u>	0.07	<u>1465.75</u>	TL
	7	15	36.10	—	—	—	0	0.77	0	0.77
	8	15	931.20	—	—	—	0	3.83	0	3.83
	9	18	3086.85	—	—	—	<u>321.75</u>	0.32	<u>321.75</u>	TL
100	0	20	706.34	—	—	—	0	0.53	0	0.53
	1	16	773.62	—	—	—	<u>546.28</u>	0.09	<u>546.28</u>	TL
	2	20	1924.02	—	—	—	0	0.31	0	0.31
	3	19	1220.14	—	—	—	0	0.85	0	0.85
	4	22	2331.66	—	—	—	0	0.30	0	0.30
	5	19	315.92	—	—	—	0	0.38	0	0.38
	6	19	4929.98	—	—	—	0	0.94	0	0.94
	7	18	2426.08	—	—	—	<u>1138.18</u>	0.20	<u>1138.18</u>	TL
	8	22	921.76	—	—	—	0	2.39	0	2.39
	9	18	2292.64	—	—	—	<u>372.58</u>	0.81	<u>372.58</u>	TL

Table C.7: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 1.0$, $RDD = 0.6$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	10	135.40	0	0.98	0	0	0.09	0	0.09
	1	11	507.40	0	7.66	0	0	0.51	0	0.51
	2	10	1498.00	0	36.66	0	0	0.29	0	0.29
	3	11	1194.60	0	60.13	0	0	0.30	0	0.30
	4	11	232.00	0	2.74	0	0	0.10	0	0.10
	5	11	787.90	0	5.06	0	0	0.10	0	0.10
	6	11	2421.10	0.90	TL	100	0.90	12.93	0.90	520.79
	7	9	424.00	0	3.30	0	0	0.13	0	0.13
	8	10	124.00	0	4.68	0	0	0.06	0	0.06
	9	11	292.70	0	3.27	0	0	0.46	0	0.46
30	0	15	1494.87	0	219.59	0	0	0.29	0	0.29
	1	16	63.87	0	10.20	0	0	0.20	0	0.20
	2	14	2428.67	461.80	TL	30.8	461.80	0.31	461.80	195.87
	3	14	424.40	0	187.76	0	0	0.35	0	0.35
	4	14	1685.67	0	12.31	0	0.33	0.05	0	4.39
	5	15	162.00	0	62.09	0	0	0.00	0	0.00
	6	14	628.13	0	95.38	0	0	0.16	0	0.16
	7	15	840.93	0	26.34	0	0	0.08	0	0.08
	8	15	54.80	0	29.05	0	0	0.34	0	0.34
	9	17	35.20	0	47.69	0	0	0.21	0	0.21
50	0	25	1505.72	NoFeasSol	TL	–	0	0.22	0	0.22
	1	25	921.68	0	302.94	0	0	0.25	0	0.25
	2	24	148.72	0	272.87	0	0	0.08	0	0.08
	3	25	1223.12	0	114.86	0	0.24	0.07	0	640.88
	4	25	2458.04	0	3340.78	0	<u>0.16</u>	0.06	<u>0.16</u>	TL
	5	25	1025.12	NoFeasSol	TL	–	0	0.19	0	0.19
	6	25	875.68	0	497.83	0	0	0.18	0	0.18
	7	25	1283.12	NoFeasSol	TL	–	0	0.17	0	0.17
	8	23	2044.44	0	699.63	0	0	0.18	0	0.18
	9	24	1194.60	NoFeasSol	TL	–	0	0.08	0	0.08
80	0	37	657.80	–	–	–	0	0.18	0	0.18
	1	40	1112.77	–	–	–	0	0.56	0	0.56
	2	35	2407.55	–	–	–	<u>0.08</u>	0.06	<u>0.08</u>	TL
	3	35	278.35	–	–	–	0	0.23	0	0.23
	4	41	237.48	–	–	–	0	0.33	0	0.33
	5	40	1037.75	–	–	–	0	0.23	0	0.23
	6	39	848.30	–	–	–	0	0.24	0	0.24
	7	37	884.45	–	–	–	0	0.13	0	0.13
	8	37	5.80	–	–	–	0	0.19	0	0.19
	9	40	2679.12	–	–	–	0	0.01	0	0.01
100	0	46	1002.02	–	–	–	0	0.22	0	0.22
	1	45	404.88	–	–	–	0	0.42	0	0.42
	2	50	1756.74	–	–	–	0	0.16	0	0.16
	3	47	3358.24	–	–	–	0	0.36	0	0.36
	4	52	2294.80	–	–	–	0	0.16	0	0.16
	5	48	349.34	–	–	–	0	0.50	0	0.50
	6	49	2401.56	–	–	–	0	0.15	0	0.15
	7	47	1523.86	–	–	–	0	0.21	0	0.21
	8	51	460.22	–	–	–	0	0.32	0	0.32
	9	47	1399.70	–	–	–	0	0.53	0	0.53

Table C.8: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 1.0$, $RDD = 0.8$)

n	Inst.#	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	8	51.60	0	9.85	0	0.10	0.13	0	2.01
	1	10	666.00	0	15.97	0	0.10	0.13	0	6.56
	2	8	1952.90	597.10	5.52	0	597.10	0.02	597.10	4.60
	3	8	1069.00	160.20	4.89	0	160.20	1.15	160.20	3.96
	4	9	865.50	0	5.03	0	0	0.22	0	0.22
	5	9	1188.40	0	7.40	0	0	0.42	0	0.42
	6	9	1521.20	266.10	40.35	0	266.10	3.59	266.10	118.48
	7	9	59.10	0	5.29	0	0	0.56	0	0.56
	8	10	59.60	0	8.79	0	0	0.12	0	0.12
	9	10	61.70	0	24.32	0	0	0.48	0	0.48
30	0	12	1183.20	0	65.02	0	0	0.21	0	0.21
	1	15	982.13	0	39.30	0	0	0.60	0	0.60
	2	11	2647.93	900.20	1414.01	0	928.07	0.34	900.20	72.45
	3	11	286.07	0	12.41	0	0	0.19	0	0.19
	4	13	224.80	0	84.60	0	0	0.00	0	0.00
	5	13	103.47	0	7.61	0	0	0.11	0	0.11
	6	12	812.40	0	131.78	0	0	0.33	0	0.33
	7	14	181.47	0	70.49	0	0	0.26	0	0.26
	8	12	194.00	0	45.36	0	0	0.18	0	0.18
	9	14	207.33	NoFeasSol	TL	–	0	0.95	0	0.95
50	0	21	772.24	NoFeasSol	3599.94	–	0	0.66	0	0.66
	1	20	51.44	NoFeasSol	TL	–	0	0.01	0	0.01
	2	20	1261.64	0	955.44	0	0	0.76	0	0.76
	3	21	2156.20	0	383.64	0	0	0.15	0	0.15
	4	22	3395.88	NoFeasSol	TL	–	0	0.17	0	0.17
	5	21	1115.76	0	3271.14	0	0	0.15	0	0.15
	6	20	961.24	0	1355.75	0	0	0.60	0	0.60
	7	21	1374.32	NoFeasSol	TL	–	0	0.08	0	0.08
	8	20	729.28	0	1212.94	0	0	0.33	0	0.33
	9	20	1136.96	0.04	TL	100	0	0.17	0	0.17
80	0	30	1278.50	–	–	–	0	0.34	0	0.34
	1	33	1619.98	–	–	–	0	0.01	0	0.01
	2	29	20.58	–	–	–	0	0.01	0	0.01
	3	29	138.45	–	–	–	0	1.53	0	1.53
	4	32	982.98	–	–	–	0	0.14	0	0.14
	5	32	998.75	–	–	–	0	0.51	0	0.51
	6	32	2742.35	–	–	–	0	0.38	0	0.38
	7	30	1186.25	–	–	–	0	0.24	0	0.24
	8	31	1321.55	–	–	–	0	0.20	0	0.20
	9	33	818.45	–	–	–	0	0.09	0	0.09
100	0	38	76.10	–	–	–	0	0.01	0	0.01
	1	36	2067.86	–	–	–	0	0.32	0	0.32
	2	40	745.16	–	–	–	0	0.01	0	0.01
	3	40	2533.58	–	–	–	0	0.12	0	0.12
	4	42	1976.78	–	–	–	0	0.01	0	0.01
	5	39	165.12	–	–	–	0	0.23	0	0.23
	6	40	4031.22	–	–	–	0	0.23	0	0.23
	7	38	137.80	–	–	–	0	0.41	0	0.41
	8	40	1595.64	–	–	–	0	0.01	0	0.01
	9	39	1440.68	–	–	–	0	0.63	0	0.63

Table C.9: Computational results for MIP-P+VI, heuristic and CP (warm-started) methods ($TF = 1.0$, $RDD = 1.0$)

n	Inst. #	$U_{\min}$	UB_{η}	MIP-P+VI			Heuristic		CP (warm-started)	
				η	Time (s)	Gap (%)	η	Time (s)	η	Time (s)
20	0	8	675.5	0	17.99	0	0	0.74	0	0.74
	1	8	55.10	0	7.14	0	0	0.36	0	0.36
	2	7	680.60	0.1	35.10	0	0.1	0.75	0.1	13.90
	3	7	1177.80	618.9	5.41	0	618.9	0.80	618.9	2.31
	4	8	556.80	0	11.58	0	0	0.67	0	0.67
	5	7	996.50	681.5	0.77	0	681.5	0.02	681.5	0.41
	6	8	1537.70	133.1	8.48	0	133.1	0.42	133.1	6.85
	7	7	80	0	18.24	0	0	0.38	0	0.38
	8	8	1069.5	0.8	2.31	0	122.1	0.02	0.8	1.33
	9	8	1681.1	674	5.83	0	674	1.11	674	8.73
30	0	9	667.4	0	7.92	0	0	0.004	0	0.004
	1	11	616.2	0	63.69	0	0	0.25	0	0.25
	2	10	2227	0	222.06	0	0	3.00	0	3.00
	3	10	350.07	0	19.72	0	0	0.45	0	0.45
	4	10	951.8	0	129.55	0	0	0.13	0	0.13
	5	11	535.2	0	61.43	0	0	0.25	0	0.25
	6	9	569.87	0	17.37	0	0	0.29	0	0.29
	7	11	84.4	0	80.81	0	0	0.10	0	0.10
	8	12	75.27	0	12.98	0	0	0.36	0	0.36
	9	12	85.33	0	231.01	0	0	0.37	0	0.37
50	0	17	764.44	0	3286.43	0	0	0.26	0	0.26
	1	16	291.16	0	96.50	0	0	0.10	0	0.10
	2	16	970.4	NoFeasSol	TL	–	0	0.15	0	0.15
	3	19	1461.32	NoFeasSol	TL	–	0	0.33	0	0.33
	4	17	1114.56	0	3307.22	0	0	0.51	0	0.51
	5	16	37.28	NoFeasSol	TL	–	0	0.32	0	0.32
	6	17	111.88	0	228.84	0	0	0.005	0	0.005
	7	16	477.4	0.04	TL	100	0	0.51	0	0.51
	8	15	445.92	0	3369.96	0	0	0.14	0	0.14
	9	17	834.56	0	3290.73	0	0	0.41	0	0.41
80	0	24	618.85	–	–	–	0	0.18	0	0.18
	1	26	894.53	–	–	–	0	0.36	0	0.36
	2	23	633.88	–	–	–	0	0.14	0	0.14
	3	24	509.95	–	–	–	0	0.36	0	0.36
	4	26	670.4	–	–	–	0	0.47	0	0.47
	5	26	2411.48	–	–	–	0	0.76	0	0.76
	6	27	2148.1	–	–	–	0	0.75	0	0.75
	7	25	201.58	–	–	–	0	0.008	0	0.008
	8	25	2731.07	–	–	–	0	2.24	0	2.24
	9	26	1935.57	–	–	–	0	0.21	0	0.21
100	0	31	274.52	–	–	–	0	0.87	0	0.87
	1	29	13.56	–	–	–	0	0.009	0	0.009
	2	31	1190.14	–	–	–	0	0.23	0	0.23
	3	32	2263.2	–	–	–	0	0.009	0	0.009
	4	37	585.62	–	–	–	0	0.36	0	0.36
	5	33	1704.5	–	–	–	0	0.55	0	0.55
	6	32	3667.08	–	–	–	<u>595.9</u>	0.10	<u>595.9</u>	TL
	7	32	554.86	–	–	–	0	0.20	0	0.20
	8	34	1173.6	–	–	–	0	0.22	0	0.22
	9	33	2019.08	–	–	–	<u>275.78</u>	0.08	<u>275.78</u>	TL

Table D.1: Results of four heuristic methods for instances with $TF = 0.6$, $RDD = 0.6$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based	EDD-based	WT-dispatch	Heuristic
				η	η	η	η
20	0	4	181.30	181.30	406.60	163.80	0.50
	1	4	975.70	975.70	1012.00	793.00	10.00
	2	4	767.80	767.80	767.80	695.20	372.10
	3	5	1020.50	1020.50	1020.50	802.30	0.10
	4	5	1186.70	1186.70	1186.70	943.80	0
	5	5	374.30	374.30	497.30	376.20	0.30
	6	5	339.40	339.40	662.40	172.00	0.20
	7	4	195.20	195.20	82.40	4.40	0.30
	8	5	623.70	623.70	623.70	439.10	1.10
	9	5	512.60	512.60	512.60	32.10	0.30
30	0	6	74.40	74.40	25.53	50.20	0.07
	1	6	48.40	48.40	32.53	147.47	0
	2	5	1113.60	1113.60	1081.00	951.13	557.80
	3	5	431.00	431.00	431.00	561.47	0.07
	4	6	460.13	460.13	382.80	274.13	72.00
	5	5	115.53	115.53	186.47	148.73	3.2
	6	5	201.13	201.13	183.20	362.53	0.2
	7	7	472.87	472.87	482.07	475.73	0
	8	6	623.00	623.00	501.80	391.87	0.27
	9	6	301.40	301.40	232.53	224.20	0
50	0	10	1050.16	1050.16	1041.68	783.16	0
	1	9	850.44	850.44	672.44	450.80	0
	2	9	823.36	823.36	772.56	273.08	0
	3	10	1268.40	1268.40	1249.32	728.08	0
	4	10	466.24	466.24	468.44	753.44	0
	5	9	1052.36	1052.36	826.24	1037.04	0
	6	10	942.96	942.96	878.28	668.24	0
	7	10	686.52	686.52	669.76	443.80	0
	8	8	103.68	103.68	139.36	29.76	0.04
	9	9	757.92	757.92	721.96	574.92	0
80	0	13	199.70	199.70	167.15	361.12	0
	1	15	715.50	715.50	660.92	551.20	0
	2	13	469.63	469.63	469.63	438.75	0
	3	13	2328.38	2328.38	2498.77	2265.30	0
	4	15	623.25	623.25	705.83	654.60	0
	5	16	585.50	585.50	354.38	459.88	0
	6	15	1115.42	1115.42	1165.90	666.08	0
	7	13	554.95	554.95	421.60	805.08	0
	8	14	466.80	466.80	496.05	266.18	0
	9	16	2653.60	2653.60	2640.25	1843.58	0
100	0	17	621.68	621.68	138.92	5.02	0
	1	16	873.26	873.26	872.74	891.68	0
	2	20	3300.06	3300.06	3200.78	2754.86	0
	3	17	2278.20	2278.20	2142.80	2119.92	4.80
	4	19	1388.00	1388.00	1399.84	1749.24	0
	5	18	998.80	998.80	696.70	1213.40	0
	6	18	2246.74	2246.74	2383.50	1343.48	4.80
	7	18	1900.78	1900.78	1799.90	1630.66	0
	8	19	285.88	285.88	404.68	227.34	0
	9	17	490.44	490.44	613.16	637.38	4.80

Table D.2: Results of four heuristic methods for instances with $TF = 0.6$, $RDD = 0.8$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	3	683.20	683.20	683.20	683.20	423.10
	1	4	2086.40	2086.40	2086.40	2086.40	2086.40
	2	3	289.90	289.90	291.00	291.00	115.00
	3	4	1460.50	1460.50	1460.50	1407.80	406.70
	4	4	83.70	83.70	83.70	260.30	38.90
	5	4	755.70	755.70	755.70	560.10	4.70
	6	3	633.70	633.70	511.70	383.20	3.20
	7	3	78.20	78.20	78.20	116.00	1.90
	8	5	929.00	929.00	929.00	617.60	16.90
	9	5	490.10	490.10	490.10	312.50	66.70
30	0	4	268.40	268.40	268.40	262.33	62.87
	1	5	523.73	523.73	570.67	369.20	0
	2	4	748.27	748.27	748.27	600.93	360.53
	3	5	349.67	349.67	490.47	557.60	0.27
	4	5	532.53	532.53	534.13	608.27	0
	5	4	159.33	159.33	159.33	288.67	0.47
	6	4	210.00	210.00	255.20	224.80	0.20
	7	5	1414.87	1414.87	1383.80	1597.53	0
	8	7	746.27	746.27	746.27	667.60	0
	9	7	1229.73	1229.73	1229.73	1104.87	5.33
50	0	8	1273.32	1273.32	1273.32	1108.04	0
	1	7	435.28	435.28	486.48	275.20	0
	2	9	351.16	351.16	367.12	309.44	0
	3	7	1036.80	1036.80	1036.80	911.00	420.08
	4	10	1156.48	1156.48	1175.76	1225.40	0
	5	9	661.12	661.12	796.00	738.80	0
	6	9	654.08	654.08	526.60	567.24	0
	7	9	656.32	656.32	630.16	277.76	0
	8	7	208.16	208.16	220.08	407.56	0.48
	9	7	1575.20	1575.20	1575.32	1292.12	0
80	0	10	635.60	635.60	662.05	578.17	344.85
	1	11	466.40	466.40	465.35	574.33	0
	2	10	256.15	256.15	183.25	43.62	0
	3	13	2982.62	2982.62	3054.40	2993.85	0
	4	13	667.25	667.25	824.33	431.75	0
	5	13	968.63	968.63	954.70	701.40	0
	6	10	1485.37	1485.37	1483.45	1426.92	900.95
	7	9	585.20	585.20	553.20	651.83	0
	8	9	639.63	639.63	622.50	527.25	0
	9	15	3762.60	3762.60	3714.85	3417.45	0
100	0	13	100.90	100.90	204.10	383.28	0
	1	11	892.58	892.58	839.76	782.44	0.02
	2	13	502.02	502.02	522.08	483.28	0
	3	13	879.64	879.64	696.04	515.44	0
	4	13	509.78	509.78	432.00	484.04	0
	5	13	324.78	324.78	344.82	613.06	0
	6	12	1780.12	1780.12	1730.24	1706.08	0
	7	17	845.82	845.82	866.12	739.20	0
	8	14	2854.38	2854.38	2722.82	2672.24	0
	9	13	2140.70	2140.70	2155.80	1858.72	0

Table D.3: Results of four heuristic methods for instances with $TF = 0.6$, $RDD = 1.0$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	4	248.10	248.10	189.30	189.30	0.20
	1	3	925.00	925.00	925.00	841.60	841.60
	2	3	289.90	289.90	291.00	291.00	115.00
	3	3	564.00	564.00	564.00	561.80	383.10
	4	3	311.20	311.20	332.50	332.50	159.50
	5	2	966.80	966.80	966.80	966.80	966.80
	6	2	303.50	303.50	175.00	303.50	0.10
	7	2	23.00	23.00	23.00	60.80	0.90
	8	4	416.20	416.20	416.20	465.00	109.00
	9	4	2239.40	2239.40	2239.40	2238.20	1205.30
30	0	2	67.27	67.27	67.27	67.27	5.13
	1	4	20.73	20.73	11.27	183.67	0.07
	2	2	1336.00	1336.00	1336.00	1336.00	1336.00
	3	3	471.80	471.80	471.80	424.80	198.87
	4	4	277.60	277.60	277.60	436.53	0
	5	3	331.13	331.13	331.13	331.13	2.07
	6	2	435.00	435.00	450.60	435.00	23.80
	7	3	335.27	335.27	335.27	335.27	24.20
	8	6	455.20	455.20	455.20	257.33	165.40
	9	6	282.13	282.13	219.47	206.80	0.13
50	0	6	862.44	862.44	828.32	762.68	0
	1	5	660.08	660.08	660.08	648.08	256.28
	2	6	539.80	539.80	579.16	497.96	0
	3	5	574.08	574.08	574.08	547.80	2.32
	4	3	1620.84	1620.84	1620.84	1620.80	1620.80
	5	4	974.56	974.56	974.56	998.76	126.16
	6	5	14.24	14.24	14.24	31.20	0.08
	7	5	51.64	51.64	51.64	23.64	0.12
	8	5	142.56	142.56	115.36	115.36	0.20
	9	4	673.76	673.76	673.76	617.12	0.52
80	0	5	853.90	853.90	940.38	909.85	609.60
	1	8	150.03	150.03	142.95	63.00	1.33
	2	8	642.28	642.28	705.12	522.88	0
	3	9	1636.88	1636.88	1657.78	1661.45	0.30
	4	5	640.18	640.18	640.17	754.90	0.10
	5	7	1338.17	1338.17	1335.55	1206.25	0
	6	6	2252.68	2252.68	2252.68	2180.15	0.05
	7	5	625.08	625.08	625.08	632.95	0.65
	8	6	316.23	316.23	288.68	366.70	0
	9	11	1058.07	1058.07	1059.50	1033.42	655.95
100	0	7	2186.16	2186.16	2186.16	2049.12	0.32
	1	6	136.64	136.64	118.86	30.68	0
	2	9	1930.78	1930.78	1939.38	1778.06	0
	3	9	449.82	449.82	502.04	682.40	0
	4	7	211.66	211.66	243.46	155.78	0
	5	7	356.26	356.26	348.64	224.64	0
	6	8	742.32	742.32	683.00	737.44	0
	7	12	806.66	806.66	773.56	862.12	0.02
	8	7	2353.96	2353.96	2328.30	2315.88	0
	9	8	2025.54	2025.54	2025.54	1986.96	0

Table D.4: Results of four heuristic methods for instances with $TF = 0.8$, $RDD = 0.6$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	8	675.50	675.50	576.50	433.50	0
	1	8	55.10	55.10	66.70	1.20	0
	2	7	680.60	680.60	680.60	888.50	0.10
	3	7	1177.80	1177.80	1228.40	1069.30	618.90
	4	8	556.80	556.80	622.20	367.10	0
	5	7	996.50	996.50	996.50	992.50	681.50
	6	8	1537.70	1537.70	1537.70	245.60	133.10
	7	7	80.00	80.00	62.30	225.70	0
	8	8	1069.50	1069.50	1069.50	614.10	122.10
	9	8	1681.10	1681.10	1849.30	1806.60	674.00
30	0	9	667.40	667.40	667.40	276.47	0
	1	11	616.20	616.20	645.40	235.80	0
	2	10	2227.00	2227.00	2227.00	1384.47	0
	3	10	350.07	350.07	274.67	91.27	0
	4	10	951.80	951.80	914.60	433.80	0
	5	11	535.20	535.20	524.13	135.93	0
	6	9	569.87	569.87	392.60	320.13	0
	7	11	84.40	84.40	125.73	312.87	0
	8	12	75.27	75.27	38.13	333.07	0
	9	12	85.33	85.33	85.33	529.40	0
50	0	17	764.44	764.44	931.44	554.88	0
	1	16	291.16	291.16	362.64	377.68	0
	2	16	970.40	970.40	709.28	1107.44	0
	3	19	1461.32	1461.32	1599.24	1125.00	0
	4	17	1114.56	1114.56	1038.32	908.80	0
	5	16	37.28	37.28	2.88	171.56	0
	6	17	111.88	111.88	508.12	242.56	0
	7	16	477.40	477.40	439.88	446.72	0
	8	15	445.92	445.92	424.64	781.36	0
	9	17	834.56	834.56	879.28	256.44	0
80	0	24	618.85	618.85	704.90	1451.35	0
	1	26	894.53	894.52	1123.35	1336.28	0
	2	23	633.88	633.88	639.52	110.78	0
	3	24	509.95	509.95	731.35	789.67	0
	4	26	670.40	670.40	513.10	915.15	0
	5	26	2411.48	2411.48	2345.20	2068.73	0
	6	27	2148.10	2148.10	2302.95	1897.85	0
	7	25	201.58	201.57	215.53	1280.15	0
	8	25	2731.07	2731.07	2567.50	1482.05	0
	9	26	1935.57	1935.57	1998.75	1204.12	0
100	0	31	274.52	274.52	332.38	170.68	0
	1	29	13.56	13.56	109.88	63.60	0
	2	31	1190.14	1190.14	1204.18	2147.66	0
	3	32	2263.20	2263.20	2204.14	1203.56	0
	4	37	585.62	585.62	589.98	305.06	0
	5	33	1704.50	1704.50	1677.30	1769.66	0
	6	32	3667.08	3667.08	3644.36	2653.22	595.90
	7	32	554.86	554.86	291.18	1179.08	0
	8	34	1173.60	1173.60	1056.26	327.44	0
	9	33	2019.08	2019.08	2081.66	1683.30	275.78

Table D.5: Results of four heuristic methods for instances with $TF = 0.8$, $RDD = 0.8$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	6	22.30	22.30	22.30	287.50	0
	1	7	1039.90	1039.90	1039.90	1125.40	0
	2	6	181.00	181.00	181.00	108.40	0.10
	3	7	1901.50	1901.50	1901.50	1564.60	543.60
	4	7	556.50	556.50	556.50	362.60	0.10
	5	6	736.60	736.60	736.60	587.60	332.80
	6	6	702.00	702.00	690.00	35.60	0.40
	7	6	257.50	257.50	370.30	458.00	0.20
	8	7	61.20	61.20	61.20	34.80	0.10
	9	7	511.20	511.20	511.20	1062.50	411.40
30	0	8	736.27	736.27	572.80	388.00	0
	1	9	1342.80	1342.80	1323.40	1031.13	0
	2	7	1127.13	1127.13	1094.53	846.27	524.67
	3	7	201.33	201.33	171.40	28.40	0
	4	8	165.53	165.53	88.20	176.07	0.67
	5	8	324.20	324.20	333.93	123.27	0
	6	7	197.80	197.80	169.67	137.33	0
	7	10	257.07	257.07	344.47	107.60	0
	8	9	622.27	622.27	543.60	267.27	0
	9	9	1030.47	1030.47	1099.33	1114.80	0
50	0	13	1247.36	1247.36	1379.24	900.96	734.64
	1	13	41.52	41.52	107.20	284.56	0
	2	13	764.80	764.80	774.16	127.92	0
	3	14	1267.96	1267.96	1224.64	513.48	0
	4	14	616.84	616.84	684.28	521.72	0
	5	13	1201.80	1201.80	1040.44	1308.24	0
	6	14	749.32	749.32	878.00	603.60	0
	7	14	202.52	202.52	213.92	35.00	0
	8	12	424.20	424.20	457.08	143.60	0
	9	13	131.44	131.44	67.80	44.48	0
80	0	18	13.53	13.53	56.80	345.60	0
	1	21	989.75	989.75	953.05	923.15	0
	2	18	837.67	837.67	836.92	741.67	0
	3	18	2566.25	2566.25	2556.40	2413.83	640.13
	4	22	109.38	109.38	393.45	845.25	0
	5	22	1471.95	1471.95	1281.70	937.80	0
	6	21	3097.38	3097.38	3041.65	2228.43	0
	7	18	869.40	869.40	790.55	1391.40	0
	8	19	1565.22	1565.22	1500.82	820.25	0
	9	21	3040.85	3040.85	2966.32	1629.28	0.03
100	0	24	684.30	684.30	542.44	543.18	0
	1	22	662.36	662.36	559.70	203.42	0
	2	27	3377.48	3377.48	3473.04	3005.16	0.02
	3	24	1141.92	1141.92	1343.52	1348.26	0
	4	27	709.78	709.78	1151.32	1469.24	0.02
	5	24	270.38	270.38	216.46	874.94	0.06
	6	25	3451.60	3451.60	3411.28	2315.22	1336.14
	7	25	250.18	250.18	286.08	156.08	0
	8	25	296.80	296.80	279.54	421.70	0
	9	24	1895.72	1895.72	1862.50	1815.42	0.06

Table D.6: Results of four heuristic methods for instances with $TF = 0.8$, $RDD = 1.0$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	4	510.30	510.30	492.80	261.70	0.80
	1	5	263.20	263.20	316.60	316.60	25.20
	2	4	730.60	730.60	730.60	693.20	2.10
	3	5	909.10	909.10	909.10	804.10	200.20
	4	5	377.80	377.80	377.80	399.10	3.20
	5	5	354.90	354.90	354.90	147.10	0
	6	5	1337.50	1337.50	1214.70	877.70	0
	7	5	310.70	310.70	310.70	348.50	0
	8	7	445.10	445.10	445.10	294.20	0
	9	5	1616.40	1616.40	1616.40	1516.70	1516.70
30	0	6	711.40	711.40	711.40	596.87	0
	1	7	660.40	660.40	755.20	871.73	0
	2	6	1268.07	1268.07	1277.13	1008.60	0
	3	5	174.80	174.80	122.60	21.60	14.80
	4	7	16.27	16.27	76.13	214.13	0
	5	5	710.53	710.53	710.53	524.00	467.73
	6	6	261.93	261.93	201.73	105.07	0
	7	7	552.87	552.87	484.47	473.60	0
	8	7	249.13	249.13	277.73	86.87	0
	9	8	1376.93	1376.93	1362.60	1249.53	0
50	0	10	1232.44	1232.44	1157.60	825.68	0
	1	10	683.28	683.28	461.72	792.76	0
	2	11	817.24	817.24	983.56	622.72	0
	3	10	1211.76	1211.76	1218.76	982.04	149.80
	4	12	545.28	545.28	515.80	117.60	0.08
	5	10	663.48	663.48	646.16	906.40	0
	6	12	1378.44	1378.44	1331.12	1087.44	0
	7	12	325.52	325.52	289.28	511.56	0
	8	9	673.56	673.56	553.88	566.36	0
	9	11	1914.20	1914.20	1851.92	1639.52	455.96
80	0	13	565.70	565.70	558.88	579.27	0.05
	1	15	1057.62	1057.62	1155.75	1415.23	0
	2	13	1075.60	1075.60	1035.90	680.23	0
	3	14	3211.32	3211.32	2946.28	3425.62	1812.58
	4	15	547.88	547.88	304.55	451.30	0
	5	17	658.75	658.75	670.95	856.50	0
	6	15	2258.15	2258.15	2279.40	1862.83	1465.75
	7	15	36.10	36.10	12.10	565.10	0
	8	15	931.20	931.20	828.83	881.17	0
	9	18	3086.85	3086.85	3071.40	2277.23	321.75
100	0	20	706.34	706.34	982.80	1081.92	0
	1	16	773.62	773.62	793.10	1189.46	546.28
	2	20	1924.02	1924.02	1888.74	1985.44	0
	3	19	1220.14	1220.14	1219.38	613.96	0
	4	22	2331.66	2331.66	2361.68	2286.82	0
	5	19	315.92	315.92	208.22	33.04	0
	6	19	4929.98	4929.98	4806.02	4360.02	0
	7	18	2426.08	2426.08	2418.16	2136.66	1138.18
	8	22	921.76	921.76	959.44	414.56	0
	9	18	2292.64	2292.64	2277.52	2055.90	372.58

Table D.7: Results of four heuristic methods for instances with $TF = 1.0$, $RDD = 0.6$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	10	135.40	135.40	27.70	277.10	0
	1	11	507.40	507.40	434.50	707.30	0
	2	10	1498.00	1498.00	1788.10	1457.70	0
	3	11	1194.60	1194.60	1194.60	1211.90	0
	4	11	232.00	232.00	205.90	553.70	0
	5	11	787.90	787.90	809.60	676.00	0
	6	11	2421.10	2421.10	2437.90	1033.00	0.90
	7	9	424.00	424.00	424.00	138.80	0
	8	10	124.00	124.00	124.00	60.00	0
	9	11	292.70	292.70	207.90	892.50	0
30	0	15	1494.87	1494.87	1494.87	974.33	0
	1	16	63.87	63.87	56.00	1618.60	0
	2	14	2428.67	2428.67	2456.13	1292.93	461.80
	3	14	424.40	424.40	464.13	400.07	0
	4	14	1685.67	1685.67	1378.93	462.67	0.33
	5	15	162.00	162.00	42.20	216.60	0
	6	14	628.13	628.13	703.20	36.53	0
	7	15	840.93	840.93	817.80	382.67	0
	8	15	54.80	54.80	138.13	694.27	0
	9	17	35.20	35.20	5.93	707.00	0
50	0	25	1505.72	1505.72	1537.80	225.96	0
	1	25	921.68	921.68	996.64	90.16	0
	2	24	148.72	148.72	202.88	166.20	0
	3	25	1223.12	1223.12	1405.12	704.44	0.24
	4	25	2458.04	2458.04	2294.88	2066.36	0.16
	5	25	1069.92	1069.92	892.12	540.68	0
	6	25	875.68	875.68	805.60	1363.44	0
	7	25	1283.12	1283.12	1472.24	1248.52	0
	8	23	2044.44	2044.44	2052.48	700.44	0
	9	24	1194.60	1194.60	1195.68	742.80	0
80	0	37	661.10	661.10	755.95	690.17	0
	1	40	1112.77	1112.77	981.88	436.68	0
	2	35	2407.55	2407.55	2394.93	565.58	0.08
	3	35	278.35	278.35	281.02	319.55	0
	4	41	237.47	237.47	143.05	166.65	0
	5	40	1037.75	1037.75	882.50	1033.20	0
	6	39	848.30	848.30	840.40	1446.27	0
	7	37	855.75	855.75	982.50	1029.43	0
	8	37	5.80	5.80	478.55	433.27	0
	9	40	2689.10	2689.10	2364.35	792.80	0
100	0	46	1002.02	1002.02	994.98	582.82	0
	1	45	404.88	404.88	374.20	1164.02	0
	2	50	1756.74	1756.74	1768.70	706.30	0
	3	47	3358.24	3358.24	3336.14	2351.22	0
	4	52	2266.42	2266.42	2109.56	737.84	0
	5	48	340.32	340.32	221.48	1050.50	0
	6	49	2401.56	2401.56	2224.34	2137.08	0
	7	47	1523.86	1523.86	980.40	549.06	0
	8	51	460.22	460.22	354.48	529.48	0
	9	47	1399.70	1399.70	1060.98	621.88	0

Table D.8: Results of four heuristic methods for instances with $TF = 1.0$, $RDD = 0.8$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	8	51.60	51.60	168.80	343.00	0.10
	1	10	666.00	666.00	1127.80	774.00	0.10
	2	8	1952.90	1952.90	1952.90	1814.70	597.10
	3	8	1069.00	1069.00	944.00	1050.60	160.20
	4	9	865.50	865.50	865.50	590.60	0
	5	9	1188.40	1188.40	1024.40	405.80	0
	6	9	1521.20	1521.20	1493.70	769.90	266.10
	7	9	59.10	59.10	59.10	284.20	0
	8	10	59.60	59.60	59.60	170.20	0
	9	10	61.70	61.70	61.70	655.50	0
30	0	12	1183.20	1183.20	1016.53	604.73	0
	1	15	982.13	982.13	1060.60	1073.00	0
	2	11	2647.93	2647.93	2692.20	1743.13	928.07
	3	11	286.07	286.07	286.07	597.47	0
	4	13	224.80	224.80	148.40	9.47	0
	5	13	103.47	103.47	95.93	267.93	0
	6	12	812.40	812.40	812.40	377.33	0
	7	14	181.47	181.47	131.13	371.40	0
	8	12	194.00	194.00	215.60	126.73	0
	9	14	207.33	207.33	345.27	142.93	0
50	0	21	772.24	772.24	824.00	260.96	0
	1	20	51.44	51.44	98.52	292.32	0
	2	20	1261.64	1261.64	1231.76	307.44	0
	3	21	2156.20	2156.20	1958.28	1450.24	0
	4	22	3395.88	3395.88	3139.56	2070.60	0
	5	21	1115.76	1115.76	1050.16	892.84	0
	6	20	961.24	961.24	1025.64	1100.60	0
	7	21	1374.32	1374.32	1388.72	999.40	0
	8	20	729.28	729.28	701.24	709.08	0
	9	20	1136.96	1136.96	1128.84	799.44	0
80	0	30	1278.50	1278.50	979.90	165.72	0
	1	33	1619.98	1619.98	1444.15	372.45	0
	2	29	20.58	20.58	121.67	940.30	0
	3	29	138.45	138.45	69.35	58.20	0
	4	32	982.98	982.98	713.55	1180.70	0
	5	32	998.75	998.75	1035.47	1576.23	0
	6	32	2742.35	2742.35	2709.45	1319.98	0
	7	30	1186.25	1186.25	1138.38	1199.97	0
	8	31	1321.55	1321.55	1479.30	366.32	0
	9	33	818.45	818.45	639.30	80.70	0
100	0	38	76.10	76.10	98.76	610.78	0
	1	36	2067.86	2067.86	2012.18	1078.88	0
	2	40	745.16	745.16	828.36	610.88	0
	3	40	2533.58	2533.58	2530.06	2622.28	0
	4	42	1976.78	1976.78	1792.04	1561.00	0
	5	39	165.12	165.12	261.16	1252.44	0
	6	40	4031.22	4031.22	3871.56	3338.52	0
	7	38	137.80	137.80	71.92	544.36	0
	8	40	1595.64	1595.64	1574.28	893.80	0
	9	39	1440.68	1440.68	1204.14	498.02	0

Table D.9: Results of four heuristic methods for instances with $TF = 1.0$, $RDD = 1.0$

n	Inst.#	$U_{\min}$	UB_{η}	Moore-based η	EDD-based η	WT-dispatch η	Heuristic η
20	0	8	675.50	675.50	576.50	433.50	0
	1	8	55.10	55.10	66.70	1.20	0
	2	7	680.60	680.60	680.60	888.50	0.10
	3	7	1177.80	1177.80	1228.40	1069.30	618.90
	4	8	556.80	556.80	622.20	367.10	0
	5	7	996.50	996.50	996.50	992.50	681.50
	6	8	1537.70	1537.70	1537.70	245.60	133.10
	7	7	80.00	80.00	62.30	225.70	0
	8	8	1069.50	1069.50	1069.50	614.10	122.10
	9	8	1681.10	1681.10	1849.30	1806.60	674.00
30	0	9	667.40	667.40	667.40	276.47	0
	1	11	616.20	616.20	645.40	235.80	0
	2	10	2227.00	2227.00	2227.00	1384.47	0
	3	10	350.07	350.07	274.67	91.27	0
	4	10	951.80	951.80	914.60	433.80	0
	5	11	535.20	535.20	524.13	135.93	0
	6	9	569.87	569.87	392.60	320.13	0
	7	11	84.40	84.40	125.73	312.87	0
	8	12	75.27	75.27	38.13	333.07	0
	9	12	85.33	85.33	85.33	529.40	0
50	0	17	764.44	764.44	931.44	554.88	0
	1	16	291.16	291.16	362.64	377.68	0
	2	16	970.40	970.40	709.28	1107.44	0
	3	19	1461.32	1461.32	1599.24	1125.00	0
	4	17	1114.56	1114.56	1038.32	908.80	0
	5	16	37.28	37.28	2.88	171.56	0
	6	17	111.88	111.88	508.12	242.56	0
	7	16	477.40	477.40	439.88	446.72	0
	8	15	445.92	445.92	424.64	781.36	0
	9	17	834.56	834.56	879.28	256.44	0
80	0	24	618.85	618.85	704.90	1451.35	0
	1	26	894.53	894.53	1123.35	1336.28	0
	2	23	633.88	633.88	639.52	110.78	0
	3	24	509.95	509.95	731.35	789.67	0
	4	26	670.40	670.40	513.10	915.15	0
	5	26	2411.48	2411.48	2345.20	2068.73	0
	6	27	2148.10	2148.10	2302.95	1897.85	0
	7	25	201.58	201.58	215.53	1280.15	0
	8	25	2731.07	2731.07	2567.50	1482.05	0
	9	26	1935.57	1935.57	1998.75	1204.12	0
100	0	31	274.52	274.52	332.38	170.68	0
	1	29	13.56	13.56	109.88	63.60	0
	2	31	1190.14	1190.14	1204.18	2147.66	0
	3	32	2263.20	2263.20	2204.14	1203.56	0
	4	37	585.62	585.62	589.98	305.06	0
	5	33	1704.50	1704.50	1677.30	1769.66	0
	6	32	3667.08	3667.08	3644.36	2653.22	595.90
	7	32	554.86	554.86	291.18	1179.08	0
	8	34	1173.60	1173.60	1056.26	327.44	0
	9	33	2019.08	2019.08	2081.66	1683.30	275.78

Table E.1: Heuristic results for $TF = 0.6$ across different RDD values

n	#	$RDD = 0.6$				$RDD = 0.8$				$RDD = 1.0$			
		U_{min}	UB_{η}	η	Time (s)	U_{min}	UB_{η}	η	Time (s)	U_{min}	UB_{η}	η	Time (s)
200	0	35	715.86	0	0.65	27	2018.20	0	5.60	20	1820.74	0	1.08
	1	32	265.45	0	0.69	23	985.44	0	0.32	12	1061.98	0	0.88
	2	34	19.71	0	4.18	28	1492.96	0	0.45	17	1021.46	0	0.58
	3	34	730.50	0	2.34	23	2581.68	0	0.71	14	627.87	0	1.62
	4	34	230.42	0	0.62	24	1027.91	0	0.01	15	535.95	0	1.03
	5	35	1976.27	0	0.39	25	122.00	0	0.65	15	987.10	0	1.86
	6	33	207.33	0	1.79	23	424.26	0	0.01	12	399.21	0	1.71
	7	34	1012.16	0	0.02	24	1384.75	172.33	0.24	12	1050.61	0	1.80
	8	34	222.52	0	0.75	23	1339.17	0	1.67	19	3135.30	0	1.06
	9	35	1989.88	0	1.62	25	4468.38	0	0.94	11	1743.87	0	1.61
Ave.		34.0	736.91	0	1.31	24.5	1588.48	17.233	1.06	14.7	1238.31	0	1.32
500	0	85	21.98	0	0.04	58	2101.78	0	1.06	29	1112.66	0	0.79
	1	81	2107.69	0	1.18	56	982.40	0	0.92	27	643.14	0	0.61
	2	85	1897.57	0	1.97	58	252.16	0	0.03	30	1628.98	0	3.49
	3	87	1302.42	0	0.04	57	1646.52	0	0.95	32	2273.96	0	4.45
	4	84	3501.98	0	2.53	56	4526.43	0	2.57	28	1938.80	0	6.13
	5	82	2559.42	0.008	0.63	56	4920.08	399.28	3.24	32	2522.88	0	3.29
	6	82	858.08	0	1.06	55	1959.16	0	4.01	27	212.94	0	3.21
	7	80	504.78	0	1.56	51	330.65	0.004	0.85	26	1509.18	0	1.78
	8	86	464.68	0	0.04	60	963.55	0	0.88	31	5596.80	0	3.17
	9	80	3423.47	0	6.81	52	921.69	0	1.61	26	380.18	0	2.50
Ave.		83.2	1664.11	0.0008	1.59	55.9	1860.44	39.9284	1.61	28.8	1681.95	0	2.94
1000	0	168	2612.38	0	2.08	112	665.04	0	1.01	58	5178.26	0	5.14
	1	165	6182.30	0	10.74	111	953.28	0	1.62	54	916.92	0	2.04
	2	167	6423.82	0.002	60.04	111	5946.76	0	6.27	56	7130.71	0	4.80
	3	167	4604.24	0	3.71	110	900.25	0	3.10	54	663.23	0	8.34
	4	172	7.16	0	0.07	114	2284.16	0.002	3.25	57	357.34	0	0.07
	5	165	646.89	0	17.99	109	1100.20	0	2.21	54	6332.86	0	5.50
	6	166	6045.12	0	7.84	109	916.57	0	0.07	56	1539.41	0	1.56
	7	166	1001.77	0	1.95	110	5319.11	1835.02	0.90	58	994.69	0	5.10
	8	162	1117.71	0	0.07	110	1069.80	0	1.04	54	4616.71	0	3.37
	9	170	4073.68	0	1.93	112	3119.68	0	1.53	60	3640.14	0	3.78
Ave.		166.8	3271.55	0.0002	10.64	110.8	2127.49	183.502	2.10	56.1	3137.03	0	3.97

Table E.2: Heuristic results for $TF = 0.8$ across different RDD values

n	#	$RDD = 0.6$				$RDD = 0.8$				$RDD = 1.0$			
		U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)
200	0	62	145.21	0.01	0.25	49	459.06	0	0.02	37	1040.37	0	0.93
	1	56	748.45	0	0.02	45	2700.58	0	0.02	35	2119.69	0	1.53
	2	61	2629.74	0	0.02	47	410.78	0	0.02	37	989.09	0	0.55
	3	63	366.39	0	1.20	47	1330.43	0.01	0.17	34	821.86	0	0.44
	4	62	765.52	0	0.02	48	797.19	0	0.36	35	525.98	0	0.68
	5	63	213.04	0	0.28	50	4.34	0	0.88	37	1167.69	0	0.46
	6	62	520.79	0	1.31	46	300.47	0	0.55	37	1436.57	0	2.42
	7	60	1360.78	0	0.55	47	936.87	0	3.20	34	2093.31	0	1.48
	8	60	981.75	0	0.57	48	2642.78	0	0.58	35	331.33	0	0.59
	9	63	559.49	0	0.73	48	2004.68	0	0.30	38	1875.22	0	0.02
Ave.		61.2	829.12	0.001	0.50	47.5	1158.62	0.001	0.61	35.9	1240.11	0	0.91
500	0	152	1613.04	0	0.04	118	864.56	0	0.04	86	2145.67	0	1.54
	1	144	5166.24	0	6.87	112	4592.97	0	0.93	83	2598.67	0	5.82
	2	153	1454.51	0	0.04	118	1341.09	0	0.67	88	2269.59	0	8.60
	3	155	1770.82	0	0.04	120	3383.97	0	0.57	88	1529.89	0	0.04
	4	149	6731.92	0	1.23	116	4618.38	0	1.48	84	3125.41	0.004	1.90
	5	151	6716.39	0	9.38	114	5245.76	0	3.29	84	4833.79	691.66	0.58
	6	149	2919.34	0	0.61	115	442.38	0	0.04	84	1913.00	0	13.72
	7	146	3332.64	0.004	1.62	112	2540.20	0	2.29	81	2705.69	0	5.65
	8	156	619.38	0	3.82	120	1480.28	0.004	0.60	87	1348.92	0	0.49
	9	145	3776.12	0	1.44	111	4738.58	0.004	1.42	80	4407.64	0	1.43
Ave.		150.0	3410.04	0.0004	2.51	115.6	2880.72	0.0008	1.13	84.5	2687.83	69.167	3.98
1000	0	301	2351.91	0	15.58	231	1268.50	0	1.24	170	306.91	0	2.51
	1	295	7266.63	0	0.07	228	4921.78	0	1.97	166	5827.53	0	2.27
	2	303	11362.00	0.008	41.94	232	8814.06	0.004	14.33	168	7312.70	0.02	1.88
	3	302	1078.50	0	2.32	232	1554.94	0	6.44	168	2362.89	0	0.07
	4	307	504.31	0	2.99	238	34.05	0	1.38	175	2811.61	0	5.28
	5	297	202.05	0	1.08	229	1708.63	0	1.19	165	2843.08	0	0.07
	6	296	4627.95	0.004	0.91	230	7034.50	0	2.45	167	3867.54	0	1.90
	7	297	5935.35	0	1.18	230	1944.92	0	13.38	167	757.79	0	3.42
	8	292	5725.79	0.002	3.25	225	1491.35	0	1.87	164	1749.62	0	1.81
	9	302	2583.98	0	3.17	235	477.12	0	30.09	171	4398.82	0	5.30
Ave.		299.2	4163.75	0.0014	7.25	230.9	2924.99	0.0004	7.34	168.1	3243.95	0.002	2.45

Table E.3: Heuristic results for $TF = 1.0$ across different RDD values

n	#	$RDD = 0.6$				$RDD = 0.8$				$RDD = 1.0$			
		U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)
200	0	96	2833.12	0	0.67	79	672.36	0	0.42	62	145.21	0.01	0.26
	1	87	3018.93	0	0.22	74	2251.07	0	0.29	56	748.45	0	0.02
	2	92	2644.09	0.04	0.17	75	493.15	0	0.02	61	2629.74	0	0.02
	3	96	216.46	0	0.84	77	987.16	0	0.40	63	366.39	0	1.30
	4	95	330.10	0	0.54	77	1853.17	0	0.43	62	765.52	0	0.02
	5	96	1253.63	0	0.30	78	132.06	0	0.79	63	213.04	0	0.29
	6	94	1677.34	0	0.20	77	686.21	0	0.28	62	520.79	0	1.35
	7	90	2252.38	0.01	0.12	75	1434.16	0	0.53	60	1360.78	0	0.56
	8	91	1287.83	0	0.38	75	618.55	0	0.02	60	981.75	0	0.58
	9	96	2531.16	0	0.02	79	2382.51	0	1.44	63	559.49	0	0.75
Ave.		93.3	1804.40	0.005	0.35	76.6	1151.34	0	0.46	61.2	829.12	0.001	0.51
500	0	230	1111.62	0.004	0.35	188	740.98	0	0.48	152	1613.04	0	0.04
	1	222	4731.66	0	1.14	181	924.47	0	0.04	144	5166.24	0	6.99
	2	235	3905.43	0	1.22	192	2616.06	0	3.03	153	1454.51	0	0.04
	3	236	2165.82	0	2.85	193	1798.80	0	0.04	155	1770.82	0	0.04
	4	234	3213.82	0	2.86	190	2465.68	0	0.04	149	6731.92	0	1.24
	5	232	4546.36	0	0.04	188	5542.44	0	0.89	151	6716.39	0	9.42
	6	230	3534.69	0.008	0.37	189	1069.83	0	0.04	149	2919.34	0	0.62
	7	224	4122.35	0	1.09	183	6889.97	0	0.88	146	3332.64	0.004	1.65
	8	236	3065.69	0.012	0.33	194	1614.45	0	2.11	156	619.38	0	3.87
	9	223	8458.24	0	2.42	180	7545.40	0	1.11	145	3776.12	0	1.49
Ave.		230.2	3885.67	0.0024	1.27	187.8	2920.71	0	0.87	150.0	3410.04	0.0004	2.34
1000	0	458	661.24	0	18.58	373	3832.46	0	1.93	301	2351.91	0	15.68
	1	456	286.89	0	3.38	372	7311.55	0.004	0.83	295	7266.63	0	0.07
	2	466	8883.72	0	2.54	379	9267.55	0.004	2.11	303	11362.00	0.008	41.53
	3	466	226.57	0	2.92	380	151.99	0	3.90	302	1078.50	0	2.37
	4	471	1529.90	0.002	1.88	384	2929.85	0	3.20	307	504.31	0	3.00
	5	456	908.53	0	0.84	372	2094.04	0	2.54	297	202.05	0	1.08
	6	453	11439.00	0	0.07	372	1942.25	0	1.33	296	4627.95	0.004	0.92
	7	458	1110.49	0	0.08	374	172.75	0.002	0.75	297	5935.35	0	1.18
	8	453	6916.54	0	1.53	366	2452.29	0	0.07	292	5725.79	0.002	3.24
	9	463	4166.69	0.002	2.79	376	3693.86	0	1.95	302	2583.98	0	3.13
Ave.		460.0	3713.06	0.0004	3.46	376.8	3484.66	0.001	1.86	299.0	4163.75	0.0014	7.22

Table E.4: Heuristic results for $TF = 0.6$ across different RDD values ($n_1 : n_2 = 1 : 5$)

n	#	$RDD = 0.6$				$RDD = 0.8$				$RDD = 1.0$			
		U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)
200	0	35	2130.01	0	0.71	27	620.56	0	0.02	20	432.91	0	2.14
	1	32	1742.81	0	0.60	23	1354.60	0	0.40	12	207.52	0	1.04
	2	34	572.37	0	7.35	28	2961.26	0	0.53	17	598.31	0	0.53
	3	34	156.79	0	2.45	23	1889.44	0	1.08	14	1145.23	0	1.23
	4	34	3337.05	0	0.53	24	815.39	0	0.38	15	3478.96	0	1.22
	5	35	1729.27	0	0.35	25	1708.99	0	0.69	15	2806.32	0	1.58
	6	33	200.33	0	2.00	23	823.11	0	0.43	12	585.22	0	1.43
	7	34	646.24	0	0.52	24	1045.99	0.006	0.25	12	187.50	0	0.93
	8	34	2770.15	0	0.57	23	823.06	0	1.05	19	3110.47	0	0.44
	9	35	743.08	0	8.21	25	1412.93	0	0.95	11	128.59	0	1.72
Ave.		34.0	1322.91	0	2.33	24.5	1345.63	0.0006	0.58	14.7	1268.10	0	1.22
500	0	85	1783.97	0	1.53	58	2020.03	0	1.03	29	2817.99	0	1.45
	1	81	785.41	0	0.04	56	231.56	0	0.03	27	1349.73	81.22	1.03
	2	85	4763.07	0	4.17	58	10236.40	0	2.07	30	119.19	0	1.47
	3	87	1863.54	0	20.36	57	1200.91	0	2.11	32	2411.39	0	1.29
	4	84	347.46	0	21.97	56	365.69	0	0.03	28	41.89	0	0.04
	5	82	2359.01	0	0.79	56	2713.65	0	0.67	32	3948.18	0	1.33
	6	82	2040.23	0	1.05	55	3283.54	0	3.73	27	894.32	0	6.76
	7	80	505.05	0	0.04	51	839.08	0.003	0.94	26	3181.80	0	1.44
	8	86	2647.70	0	1.57	60	1418.90	0	0.55	31	6452.57	0.003	6.19
	9	80	1962.66	0	1.77	52	3059.35	0	7.08	26	209.43	0	0.03
Ave.		83.2	1905.71	0	5.33	55.9	2633.01	0.0003	1.82	28.8	1952.55	8.1223	2.10
1000	0	168	3302.27	0	4.00	112	4225.08	0.006	1.00	58	4677.90	0	5.25
	1	165	23.37	0	1.85	111	4960.82	885.644	8.06	54	13558.90	0	7.54
	2	167	3602.69	0	2.56	111	1264.87	0	0.07	56	4106.57	0	16.07
	3	167	5512.55	0	1.97	110	1859.89	0	2.87	54	194.52	0	5.29
	4	172	1507.39	0	22.24	114	4171.85	0	2.09	57	5192.94	3358.73	1.94
	5	165	7598.52	0	18.18	109	6804.56	0	4.71	54	7975.43	0	3.72
	6	166	2768.62	0	24.58	109	2638.05	0	14.37	56	597.80	0	2.79
	7	166	442.36	0	1.38	110	2192.30	0.005	0.90	58	4788.60	0	3.03
	8	162	3877.57	0	0.07	110	3156.57	0.001	1.91	54	2907.18	0	6.95
	9	170	3069.50	0	4.90	112	1617.53	0	0.07	60	919.06	0	7.27
Ave.		166.8	3270.44	0	8.17	110.8	3189.15	88.5655	3.60	56.1	3991.89	335.873	5.98

Table E.5: Heuristic results for $TF = 0.8$ across different RDD values ($n_1 : n_2 = 1 : 5$)

n	#	$RDD = 0.6$				$RDD = 0.8$				$RDD = 1.0$			
		U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)
200	0	62	1386.09	0	0.46	49	182.08	0	0.72	37	2291.41	0	0.02
	1	56	31.64	0	0.39	45	2975.36	0	0.30	35	1415.61	0	0.88
	2	61	60.69	0	0.02	47	576.00	0	0.45	37	938.71	0	1.63
	3	63	1667.99	0	0.66	47	303.38	0.025	0.16	34	553.00	0	0.45
	4	62	4038.38	0	0.27	48	6065.82	0	1.92	35	256.00	0	1.38
	5	63	635.41	0	0.28	50	2036.97	0	1.12	37	177.56	0	0.34
	6	62	19.56	0	0.02	46	1089.35	0	2.40	37	447.58	0	2.51
	7	60	3235.24	0	3.69	47	804.95	0	0.02	34	406.71	0	1.29
	8	60	2297.04	0	0.38	48	4232.73	0	1.50	35	2833.44	396.84	0.73
	9	63	2028.04	0	0.90	48	2309.17	0.0125	0.18	38	495.61	0	0.41
Ave.		61.2	1516.00	0	0.71	47.5	2057.98	0.0038	0.88	35.9	981.56	39.684	0.96
500	0	152	4831.73	0	1.28	118	4375.04	0	0.04	86	4985.05	0	3.42
	1	144	1354.61	0	1.92	112	1656.37	0	0.04	83	1976.90	0	2.61
	2	153	7315.93	0	0.61	118	9256.62	0	9.38	88	5861.27	0	13.30
	3	155	647.18	0	0.04	120	1102.64	0	0.78	88	2305.14	0	0.94
	4	149	4723.61	0	2.84	116	3921.89	0	13.14	84	2165.14	0	1.59
	5	151	5572.14	0	0.04	114	5560.35	0	0.04	84	4599.06	1227.47	0.69
	6	149	1871.32	0.0025	0.50	115	6120.15	0.005	0.42	84	3019.15	0	3.96
	7	146	3163.61	0	1.57	112	180.94	0	0.66	81	2242.25	0	4.73
	8	156	4912.69	0	4.23	120	3459.46	0	0.78	87	1539.44	0.0025	0.51
	9	145	395.37	0	0.74	111	91.76	0	0.04	80	1532.62	0	3.67
Ave.		150.0	3278.86	0.0003	1.18	115.6	3572.52	0.0005	2.53	84.5	3022.60	122.747	3.54
1000	0	301	11210.20	0	13.74	231	8025.85	0.015	1.14	170	5790.45	0	1.59
	1	295	8807.33	0	6.69	228	30.66	0	6.10	166	2166.64	0.00125	3.56
	2	303	1939.56	0	4.45	232	1490.82	0	2.64	168	3203.61	0	9.20
	3	302	761.65	0	1.25	232	2372.58	0	1.70	168	3591.94	0	0.07
	4	307	4002.19	0.00125	4.38	238	2410.14	0.00125	4.26	175	3819.72	0	12.12
	5	297	10041.60	0.00125	5.29	229	13831.40	0.00125	60.02	165	11600.00	1905.56	3.09
	6	296	3767.62	0.005	0.93	230	4048.34	0	22.85	167	931.34	0	8.43
	7	297	8194.21	0.0025	1.26	230	4666.18	0	3.70	167	1233.46	0	3.34
	8	292	8587.46	0.02625	1.02	225	7222.26	0	3.43	164	5960.34	0	1.49
	9	302	583.58	0	1.63	235	59.24	0.00125	1.18	171	2329.50	0	12.40
Ave.		299.2	5789.54	0.0036	3.98	231.8	4415.75	0.0019	10.60	168.1	3662.76	190.556	5.53

Table E.6: Heuristic results for $TF = 1.0$ across different RDD values ($n_1 : n_2 = 1 : 5$)

n	#	$RDD = 0.6$				$RDD = 0.8$				$RDD = 1.0$			
		U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)	U_{min}	UB_η	η	Time (s)
200	0	96	291.89	0	1.29	79	1007.27	0	0.45	62	1386.09	0	0.43
	1	87	5785.86	0	0.21	74	2586.96	0.00625	0.18	56	31.64	0	0.39
	2	92	517.80	0	0.20	75	1987.42	0	0.02	61	60.69	0	0.02
	3	96	1690.94	0	0.28	77	2387.26	0	0.57	63	1667.99	0	0.65
	4	95	3120.33	0	0.02	77	2102.45	0	0.57	62	4038.38	0	0.28
	5	96	384.98	0	0.30	78	2616.61	0	1.12	63	635.41	0	0.33
	6	94	588.08	0	0.20	77	1437.14	0	0.33	62	19.56	0	0.02
	7	90	2936.19	0.0375	0.18	75	3067.81	0	0.44	60	3235.24	0	3.86
	8	91	4726.73	0	0.02	75	3525.73	0	0.99	60	2297.04	0	0.39
	9	96	3199.62	0	0.28	79	2374.36	0	2.26	63	2028.04	0	0.90
Ave.		93.3	1864.14	0.0038	0.30	76.6	2309.30	0.0006	0.68	61.2	1516.00	0	0.73
500	0	230	3478.73	0.02	0.40	188	2757.69	0.0075	0.46	152	4831.73	0	1.25
	1	222	1102.90	0.005	0.76	181	3713.46	0.0025	0.38	144	1354.61	0	1.96
	2	235	5318.88	0	2.69	192	10244.20	0	1.88	153	7315.93	0	0.60
	3	236	4633.58	0	1.22	193	4259.48	0	0.65	155	647.18	0	0.04
	4	234	3812.79	0	2.60	190	1255.46	0	1.03	149	4723.61	0	3.03
	5	232	5245.17	0	1.79	188	5087.60	0	2.34	151	5572.14	0	0.04
	6	230	1380.71	0.0025	0.47	189	2795.76	0	1.04	149	1871.32	0.0025	0.51
	7	224	49.47	0	0.85	183	1121.68	0	0.80	146	3163.61	0	1.63
	8	236	2627.22	0	0.04	194	4979.57	0	0.04	156	4912.69	0	4.49
	9	223	7323.46	0	0.04	180	5000.88	0	3.12	145	395.37	0	0.75
Ave.		230.2	3397.29	0.0028	1.09	187.8	4116.08	0.0010	1.17	150.0	3278.86	0.0003	1.43
1000	0	458	5444.45	0	1.30	373	6737.38	0.00625	2.07	301	11210.20	0	13.26
	1	456	5648.45	0	1.89	372	7622.89	0.00125	0.96	295	8807.33	0	6.44
	2	466	1254.57	0.00125	5.73	379	1884.66	0	2.87	303	1939.56	0	4.34
	3	466	2332.91	0	2.80	380	117.02	0.00125	4.10	302	761.65	0	1.19
	4	471	18964.80	0.00875	1.20	384	11441.50	0.02125	2.16	307	4002.19	0.00125	4.33
	5	456	11591.20	0.00625	2.06	372	7500.56	0	6.96	297	10041.60	0.00125	5.10
	6	453	15783.50	0	1.94	372	3479.71	0	4.11	296	3767.62	0.005	0.92
	7	458	5804.49	0.00125	1.72	374	4279.54	0	0.07	297	8194.21	0.0025	1.25
	8	453	348.08	0	5.23	366	3473.20	0	0.07	292	8587.46	0.02625	1.01
	9	463	4581.43	0.00125	3.35	376	2735.45	0.00125	2.13	302	583.58	0	1.59
Ave.		460.0	6575.39	0.0021	2.62	374.8	4527.19	0.0031	2.55	299.2	5789.54	0.0036	3.84